



# Supplementary Materials for

## **Metal-free efficient photocatalyst for stable visible water splitting via a two-electron pathway**

Juan Liu, Yang Liu, Naiyun Liu, Yuzhi Han, Xing Zhang, Hui Huang, Yeshayahu Lifshitz,\*  
Shuit-Tong Lee,\* Jun Zhong, Zhenhui Kang\*

\*Corresponding author. E-mail: zhkang@suda.edu.cn (Z.K.); apannale@suda.edu.cn (S.-T.L.); shayli@tx.technion.ac.il (Y.L.)

Published 27 February 2015, *Science* **347**, 970 (2015)  
DOI: 10.1126/science.aaa3145

### **This PDF file includes:**

Materials and Methods

Figs. S1 to S36

Tables S1 to S3

References

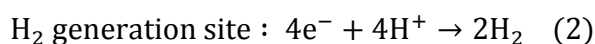
# Metal-free efficient photocatalyst for stable visible water splitting via a two-electron pathway

## Supplementary Materials

### 1. Analysis of the water splitting process.

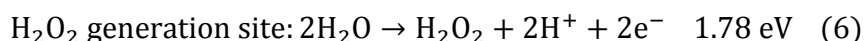
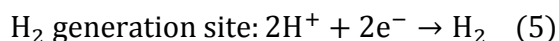
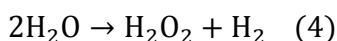
In a photocatalytic water splitting process the catalyst absorbs photons and generates electron and holes which are separated. The electrons move to the hydrogen generation surface sites and the holes move to the oxygen generation surface sites.

The typical concerted four-electron process for water splitting is:

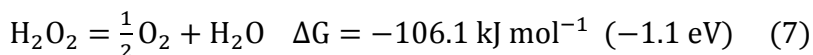


While the stepwise  $2\text{e}^-/2\text{e}^-$  two-step pathway is as follows:

First step, production of hydrogen and hydrogen peroxide from water:



Second step, disproportionation of hydrogen peroxide releases energy:



As shown above, the concerted four-electron oxidation of water to oxygen requires an energy of 1.23 eV per  $\text{e}^-$  for the chemical reactions shown in equation (1) and (3). The two-electron oxidation path forming  $\text{H}_2\text{O}_2$  from  $\text{H}_2\text{O}$  requires a larger energy of 1.78 eV per  $\text{e}^-$  for the chemical reactions shown in equation (4) and (6). In practical applications, about 1.5-1.7 eV is required for reaction (1) due to energy barriers (over potentials), whereas the actual energy needed for reaction (4) is about 2.0 eV (43). The four-electron water splitting reaction is thus thermodynamically favorable with respect to the two-electron oxidation path forming  $\text{H}_2\text{O}_2$ . If however the catalyst induced water splitting forming  $\text{H}_2\text{O}_2$  (reaction 4) is kinetically favored with respect to the four-electron process, it would consume the necessary electrons/holes needed for the four-electron process which is suppressed. The second step of the  $2\text{e}^-/2\text{e}^-$  two-step pathway ( $\text{H}_2\text{O}_2$  disproportionation) is exothermic (reaction 7), but is slow (without a second-step catalyst), and  $\text{H}_2\text{O}_2$  may poison the catalyst used for its generation. If hydrogen peroxide is however subsequently decomposed by an appropriate catalyst, the stepwise  $2\text{e}^-/2\text{e}^-$  two-step pathway may represent an optimum approach between thermodynamic and kinetic constraints for water splitting.

### 2. Experimental section.

#### 2.1 Preparation of CDots.

CDots were synthesized by the typical electrochemical method (37). Two graphite rods (99.99%, Alfa Aesar Co. Ltd.) were inserted into ultrapure water serving as anode and cathode, and a bias of 30 V was applied between the two electrodes using a direct current power supply. After a

continuous stirring for 120 h, a dark yellow solution was formed in the reaction cell. The dark yellow solution was then dried. The powder formed was weighted and added to ultrapure water to form a solution of 4 mg/L. The next step, preceding the fabrication of the CDots-C<sub>3</sub>N<sub>4</sub> composite, involved ammonia treatment. 30 mL of ammonia was mixed with 30 mL of the CDots 4 mg/L solution and stirred to a homogeneous solution. The mixed solution was then transferred into a 100 mL Teflon stainless steel autoclave at 150 °C for 7 h. The yellow solution was then collected by centrifugation and opened to air at room temperature removing the excess ammonia and achieving pH=7. Afterwards the CDots solution was dried at 60 °C to a powder which was weighted again.

## **2.2 Preparation of the C<sub>3</sub>N<sub>4</sub> and the CDots-C<sub>3</sub>N<sub>4</sub> photocatalysts.**

Typically, 10 g of urea powder was put into an alumina crucible with a cover and then heated to 550 °C at a rate of 0.5 °C/min in a muffle furnace and maintained at this temperature for 3 h. C<sub>3</sub>N<sub>4</sub> was obtained after cooling down to room temperature. For preparation of CDots-C<sub>3</sub>N<sub>4</sub>, 10 g of urea powder was added to a 5 mL 2 mg/L CDots solution. Then the mixture was put into an alumina crucible with a cover and heated to 550 °C at a rate of 0.5 °C/min in a muffle furnace and maintained at this temperature for 3 h. The resultant powder (containing 0.6 g of C<sub>3</sub>N<sub>4</sub>) was cooled to room temperature, washed with ultrapure water, collected by filtration and finally dried at room temperature. The CDots concentration obtained for a 2 mg/L CDots solution was  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub> (the concentration used for many of the characterizations). Different CDots concentrations are obtained by changing the CDots solution used.

## **2.3 Photocatalyst Characterization.**

The crystal structure of the resultant products was characterized by X-ray diffraction (XRD) using an X'Pert-ProMPD (Holand) D/max-γAX-ray diffractometer with Cu Kα radiation ( $\lambda=0.154178$  nm). Scanning electron microscopy (SEM) images and energy dispersive X-ray (EDX) spectroscopy were performed by a FEI-quanta 200 scanning electron microscope with an acceleration voltage of 20 kV. Transmission electron microscope (TEM) and high-resolution TEM (HRTEM) images were obtained with a FEI-Tecna F20 (200 kV), respectively. The TEM samples were prepared by dropping the solution onto a copper grid with polyvinyl supporting film and dried in air. Raman spectra were collected using an HR 800 Raman spectroscope (J Y, France) equipped with a synapse CCD detector and a confocal Olympus microscope. The spectrograph used 600 g mm<sup>-1</sup> gratings and a 633 nm He-Ne laser was utilized for excitation. Room temperature UV-Vis absorption was recorded using a Lambda 750 (Perkin Elmer) spectrophotometer in the wavelength range of 300–800 nm. The Fourier transform infrared (FTIR) spectrum is recorded on a FTIR spectrometer (Spectrum One, Perkin Elmer) using a standard KBr pellet technique. X-ray photoelectron spectroscopy (XPS) was obtained by using a KRATOS Axis ultra-DLD X-ray photoelectron spectrometer with a monochromatised Mg Kα X-ray source ( $h\nu = 1283.3$  eV). Ultraviolet photoelectron spectroscopy (UPS) measurements were performed with an unfiltered HeI (21.22 eV) gas discharge lamp and a total instrumental energy resolution of 100 meV. Atomic force microscopy (AFM) was performed by means of a Veeco DI Nanoscope Multi Mode V system. X-ray absorption near edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) data were collected on beamline 14 W at the Shanghai Synchrotron Radiation Facility. The electro-catalysis actions were tested by a Model CHI 660C workstation (CH Instruments, Chenhua, Shanghai, China).

## **2.4 Photocatalysis activity test.**

The photocatalytic reactions of the composite photocatalysts were carried out in an outer irradiation-type photoreactor (Pyrex glass) connected to a closed gas-circulation system. Approximately 0.080 g of the photocatalysts were dispersed by a magnetic stirrer in 150 mL ultrapure water. The suspension is thoroughly degassed to remove air and irradiated using a 300 W Xe-lamp (PLS-SXE 300, Beijing Trusttech Co. Ltd, China). A cutoff filter (Kenko L-42) was employed to achieve visible-light ( $\lambda > 420$  nm) irradiation. The photocatalytic H<sub>2</sub> evolution rate was analyzed using an online SP-6890 gas chromatograph (GC, TCD detector, 5 Å° molecular sieve columns and N<sub>2</sub> carrier). Magnetic stirring (400 rpm) was used during the water splitting experiment to ensure homogeneity of the suspension and to eliminate sedimentation. The catalyst was used in different ways: (1) continuous water splitting for up to 45 days, (2) 1 day cycles after each the catalyst was recycled by 60 °C drying in an oven, weighted and reused. In different experiments a different number of cycles up to 200 were performed.

## **2.5 Electrocatalytic activity measurements.**

The electrocatalytic activities of C<sub>3</sub>N<sub>4</sub> and CDots-C<sub>3</sub>N<sub>4</sub> for H<sub>2</sub>O<sub>2</sub> decomposition were tested in a conventional three-electrode electrochemical cell with a platinum wire as the auxiliary electrode and a saturated calomel electrode (SCE, saturated KCl) as the reference electrode. The working electrodes were: (1) bare glassy carbon (GC), (2) C<sub>3</sub>N<sub>4</sub>- or (3) CDots-C<sub>3</sub>N<sub>4</sub>-modified GC electrode with the bottom facing the incident light. A 300 W Xe lamp was used as the light source with a visible light-cutoff filter to achieve  $\lambda > 420$  nm to get visible light and positioned 3 cm away from the photoelectrochemical cell.

To study the catalase-like property of CDots, the cyclic voltammograms (CVs) of the C<sub>3</sub>N<sub>4</sub>- and CDots-C<sub>3</sub>N<sub>4</sub>-modified GC electrodes were obtained in a 0.05 M phosphate buffer solution (PBS, pH 7) in N<sub>2</sub>-saturated ultrapure water with light irradiation of  $\lambda > 420$  nm (scan rate: 50 mV s<sup>-1</sup>).

## **2.6 Scanning electrochemical microscope measurements.**

### **2.6.1 Determination of the onset potential of electro-catalytic water oxidation reaction over the catalyst.**

The onset potential of electro-catalytic water oxidation reaction over CDots-C<sub>3</sub>N<sub>4</sub> on fluorine-doped tin oxide (FTO) was assessed by scanning electrochemical microscopy (SECM). A C<sub>3</sub>N<sub>4</sub> spot is deposited on FTO. Briefly, a single drop of about 1 mm in diameter of an aqueous solution of the catalyst was deposited on FTO using a micropipet. The drop was dried at 45 °C in air for 1 h. A CHI model 920c scanning electrochemical microscope (CH Instruments, Austin, TX) was used to control the tip potentials, obtain the approach curves, and monitor the tip-to-substrate distance. The SECM tip made of a Pt disk (diameter = 25 μm) sealed in a glass capillary, SCE, and Pt wire were used as a working, reference, and counter electrodes, respectively. Prior to the experiments, the tip was manually polished with 0.05 μm alumina powder, and the surface quality of the tip was monitored under an optical microscope. The electrochemical performance of the tip was checked periodically with the help of steady-state voltammograms. All the electrochemical measurements were carried out in phosphate buffer (pH = 7) solution. The tip was biased at -0.3 V vs. SCE.

### **2.6.2 Determination of H<sub>2</sub>O<sub>2</sub> diffusion layer on C<sub>3</sub>N<sub>4</sub>.**

The C<sub>3</sub>N<sub>4</sub> spot about 1 mm in diameter was deposited on FTO (fluorine doped tin oxide). The generation of H<sub>2</sub>O<sub>2</sub> was detected by a Pt tip. The tip potential was held at 0.9 V vs SCE to detect H<sub>2</sub>O<sub>2</sub> in a phosphate buffer (pH = 7) solution. No voltage was applied on the generator FTO. The

incident light was irradiating from the bottom of the FTO by an inverted Xe lamp optical fiber. The H<sub>2</sub>O<sub>2</sub> detection current was measured along the X-axis direction at various Z positions of the detector (50, 75, 150...1000 μm above the top of the generator).

## 2.7 Photo-electrochemical measurements.

### 2.7.1 Poisoning of the C<sub>3</sub>N<sub>4</sub> surface by H<sub>2</sub>O<sub>2</sub>.

To check the photoinduced current corresponding to the generation of H<sub>2</sub>O<sub>2</sub>, 10 mg catalyst (C<sub>3</sub>N<sub>4</sub> or CDots-C<sub>3</sub>N<sub>4</sub>) was dispersed in 100 mL water. 20 μL of the suspension made was loaded on a glassy carbon electrode as a working electrode. SCE and Pt wire were used as the reference and counter electrodes, respectively. The electrochemical experiments were carried out in the current-time mode. The potential was set to null the current under dark conditions. The test was carried out in ultrapure water under 300 W Xenon lamp irradiation (λ>420 nm). The photo-generated currents were recorded by an electrochemical workstation (CHI 660e, CH Instrument, Shanghai, China).

### 2.7.2 Determination of electron transfer number.

The electron transfer number was studied by rotating Pt disk-Pt ring electrodes (RRDE). The C<sub>3</sub>N<sub>4</sub>- and CDots-C<sub>3</sub>N<sub>4</sub> modified GC electrodes at a rotating speed of 1600 rpm in N<sub>2</sub>-saturated ultrapure water. The disk potential was set at different potential (0.4, 0.6, 0.8 V vs. SCE) under different light intensity irradiation (λ>420 nm). The ring potential was set at 0.9 V (or -0.3V) vs. SCE.

## 2.8 H<sub>2</sub><sup>18</sup>O water splitting experiment.

Water splitting was performed using H<sub>2</sub><sup>18</sup>O instead of natural water to verify that O<sub>2</sub> was solely released from water splitting and no other source. Approximately 0.016 g CDots-C<sub>3</sub>N<sub>4</sub> (4.8 × 10<sup>-3</sup> g<sub>CDots</sub>/g<sub>catalyst</sub>) were dispersed by a magnetic stirrer in 30 mL H<sub>2</sub><sup>18</sup>O. The suspension was thoroughly degassed to remove air and irradiated using a 300 W Xe-lamp. A cutoff filter (Kenko L-42) was employed to achieve visible-light (λ>420 nm) irradiation. The reaction products were detected by gas chromatography-mass spectrometry (GC-MS, an Agilent Technologies 5975C Mass Selective Detector operating in the electron impact ionization mode was used).

## 3. Quantum efficiency (QE) and solar energy conversion calculations.

### 3.1 Quantum efficiency calculations.

In the following we describe the QE determination at λ<sub>0</sub>=420 nm for CDots-C<sub>3</sub>N<sub>4</sub> with a CDots concentration of 1.6 × 10<sup>-5</sup> g<sub>CDots</sub>/g<sub>catalyst</sub>. The catalyst solution was irradiated by a 300W Xe lamp applying a λ<sub>0</sub>±20 nm band-pass filter for 12 hours. The average intensity of irradiation was determined to be 2.84 mW·cm<sup>-2</sup> by an ILT 950 spectroradiometer (International Light Technologies) and the irradiation area was 4.27 cm<sup>2</sup>. The number of incident photons (N) is 1.11 × 10<sup>21</sup> as calculated by equation (8). The amount of H<sub>2</sub> molecules generated in 12 hours was 79.4 μmol (Fig. S20). The quantum efficiency is calculated from equation (9).

$$N = \frac{E\lambda}{hc} = \frac{2.84 \times 10^{-3} \times 4.27 \times 3600 \times 12 \times 420 \times 10^{-9}}{6.626 \times 10^{-34} \times 3 \times 10^8} = 1.11 \times 10^{21} \quad (8)$$

So,

$$QE = \frac{2 \times \text{the number of evolved H}_2 \text{ molecules}}{\text{the number of incident photons}} \times 100\%$$

$$= \frac{2 \times 6.02 \times 10^{23} \times 79.4 \times 10^{-6}}{1.11 \times 10^{21}} \times 100\% = 8.6\% \quad (9)$$

### 3.2 Solar to hydrogen (STH) conversion efficiency calculations from solar simulator measurements.

The solar energy conversion was evaluated by using AM 1.5G solar simulator as the light source with CDots-C<sub>3</sub>N<sub>4</sub> (4.8×10<sup>-3</sup> g<sub>CDots</sub>/g<sub>catalyst</sub>) as the catalyst (80 mg catalyst in 150 mL water). After 6 h of illumination, the total incident power over the 9 cm<sup>2</sup> irradiation area was 0.63 W, so that the total input energy in 6 hours was

$$E_{\text{Solar}} = 1.36 \times 10^4 \text{ J.}$$

During the photocatalytic reaction, 1150 μmol H<sub>2</sub> was detected by gas chromatography (GC), which indicated that the energy generated by water splitting is E<sub>F</sub> = 274 J

$$E_F = 1150 \times 10^{-6} \times 6.02 \times 10^{23} \times 2.46 \times 1.609 \times 10^{-19}; 2.46 \text{ eV is the free energy of water splitting.}$$

The “solar-to-hydrogen” conversion efficiency (STH) of CDots-C<sub>3</sub>N<sub>4</sub> with the higher concentration of CDots was determined to be:

STH = (Energy of generation of hydrogen by water splitting) / (Solar energy irradiating the reaction cell).

$$STH = E_F / E_{\text{Solar}} = 274 \text{ J} / 1.36 \times 10^4 \text{ J} = 2.0\%$$

### 3.3. STH calculations from the QE data.

The STH was independently calculated from the QE data.

$$STH(\%) = \frac{\text{Energy of generated H}_2}{\text{Solar energy irradiating the solution}} = \frac{E_{\text{H}_2}}{E_{\text{solar}}}$$

$$E_{\text{H}_2} = \sum_i \frac{n_{\text{photon},i} \times 1.23 \text{ eV} \times 1.609 \times 10^{-19} \text{ J/eV}}{E_{\text{solar}}} \times QE(\lambda_i)$$

$$n_{\text{photon},i} = \frac{p_i \times E_{\text{solar}} \times \lambda_i}{hc} = \frac{p_i \times E_{\text{solar}} \times \lambda_i}{6.626 \times 10^{-34} \times 3 \times 10^8}$$

$$STH(\%) = \sum_i \frac{p_i \times 1.23 \times 1.609 \times 10^{-19} \times \lambda_i}{19.89 \times 10^{-26}} \times QE(\lambda_i)$$

$$STH(\%) = \sum_i 0.996 \times 10^6 \times p_i \times \lambda_i \times QE(\lambda_i)$$

here,  $n_{\text{photon},i}$  - number of photons in interval  $i$ .  $p_i$  - fraction of solar power in interval  $i$ . The spectral power distribution was that of AM 1.5G.  $QE(\lambda_i)$  is the quantum efficiency of the  $i$ -th interval in percentage.

The entire solar spectrum was divided into eight intervals ( $i=1, 2, 3 \dots 8$ ). The fraction of solar power in interval  $i$  calculated by the area of interval  $i$  of the spectrum divided by the entire area of the solar spectrum distribution AM 1.5 G. The incident solar power is  $1.36 \times 10^4$  J.  $\lambda_i$  is the average wavelength of interval  $i$ .

So, STH calculated from the QE values is 1.778%.

| i                    | Range (nm) | $\lambda_i$ (nm) | QE (%) | $p_i$ (%) | STH <sub>i</sub> |
|----------------------|------------|------------------|--------|-----------|------------------|
| 1                    | 280-400    | 340              | 18.2   | 7.63      | 0.4703           |
| 2                    | 400-440    | 420              | 15.9   | 5.08      | 0.3379           |
| 3                    | 440-480    | 460              | 13.2   | 5.85      | 0.3537           |
| 4                    | 480-520    | 500              | 7.44   | 5.55      | 0.2056           |
| 5                    | 520-560    | 540              | 6.33   | 5.44      | 0.1852           |
| 6                    | 565-595    | 580              | 6.29   | 4         | 0.1453           |
| 7                    | 590-610    | 600              | 4.42   | 2.6       | 0.0687           |
| 8                    | 610-650    | 630              | 0.36   | 4.86      | 0.0110           |
| 9                    | 620-680    | 650              | 0      | 6.99      | 0                |
| 10                   | 680-4000   | 2340             | 0      | 55.59     | 0                |
| <b>Total STH (%)</b> |            |                  |        |           | <b>1.778%</b>    |

#### 4. Electrochemical measurements of the catalyst activity of CDots for H<sub>2</sub>O<sub>2</sub> decomposition.

To further prove the catalyst activity for H<sub>2</sub>O<sub>2</sub> decomposition, the electrochemical behavior of electrodes made of different catalysts was measured using cyclic voltammetry in 0.05 M (pH=7) phosphate buffered 25 mM H<sub>2</sub>O<sub>2</sub> solution under darkness. The electrochemical tests were carried out in a typical three-electrode system with the working electrode of either (1) bare glassy carbon (GC) (black trace), CDots-(green trace), or (2) C<sub>3</sub>N<sub>4</sub>-(red trace) or (3) CDots-C<sub>3</sub>N<sub>4</sub>-(blue trace) modified GC electrodes, respectively. Platinum wire served as the counter electrode, and saturated calomel electrode (SCE) as the reference electrode. Fig. S21 (and inset) clearly shows that the CDots-C<sub>3</sub>N<sub>4</sub>- and CDots-modified electrodes exhibited the most negative onset potential and the largest current at -0.90 V vs. SCE, indicating that both the CDots-C<sub>3</sub>N<sub>4</sub> composite and the pure CDots have a strong catalytic activity toward H<sub>2</sub>O<sub>2</sub> decomposition. The catalytic activity is attributed to the presence of CDots (the catalytic properties of which are shown in the green trace in Fig. S21) in the CDots-C<sub>3</sub>N<sub>4</sub> composite catalyst.

To study the catalase-like activity of CDots, the electrochemical tests were performed in pure water under light irradiation of  $\lambda > 420$  nm, and the measured cyclic voltammograms (CVs) are shown in Fig. S22. The black line exhibits a typical CV for C<sub>3</sub>N<sub>4</sub> (with an oxidation peak at 0.51 V vs. SCE and reduction peak at -0.13 V vs. SCE). But for CDots-C<sub>3</sub>N<sub>4</sub> (blue trace), distinctive oxidation and reduction peaks ( $I_a$  and  $I_b$ ) are observed on either side of the cycle, probably due to the oxygen groups associated with surface functionalities. In addition, a considerable current is observed at -0.30 V vs. SCE, which is attributed to H<sub>2</sub>O<sub>2</sub> reduction. A control experiment is carried out using C<sub>3</sub>N<sub>4</sub>-modified GC electrode in N<sub>2</sub>-saturated ultrapure water containing 1 mg/50 mL catalase (an enzyme catalyst for H<sub>2</sub>O<sub>2</sub> reduction), showing the reduction peak of H<sub>2</sub>O<sub>2</sub> at the same potential of -0.30 V vs. SCE (Fig. S22A green line). From the above CV experiments, we confirm that CDots-C<sub>3</sub>N<sub>4</sub> can efficiently catalyze H<sub>2</sub>O<sub>2</sub> decomposition with catalase-like activity.

## 5. Photoelectrochemical analysis of the poisoning effect of H<sub>2</sub>O<sub>2</sub> on C<sub>3</sub>N<sub>4</sub> surface.

We used a photo-electro-chemical method to evaluate the photocatalytic activity of C<sub>3</sub>N<sub>4</sub> and CDots-C<sub>3</sub>N<sub>4</sub>, and get the some information about the H<sub>2</sub>O<sub>2</sub> poisoning of C<sub>3</sub>N<sub>4</sub> (by adsorption to the C<sub>3</sub>N<sub>4</sub> surface and an associated decrease of the H<sub>2</sub>O<sub>2</sub> generation rate with time).

10 mg catalysts (C<sub>3</sub>N<sub>4</sub> or CDots-C<sub>3</sub>N<sub>4</sub>) were dispersed in 100 mL ultrapure water. 20  $\mu$ L of the suspension was loaded on a glassy carbon electrode serving as a working electrode. SCE and Pt wire were used as the reference and counter electrodes, respectively. The electrochemical experiments were carried out in the current-time mode. The potential was set at the value which nulls the current under darkness. The test was carried out in ultrapure water under 300 W Xenon lamp irradiation with  $\lambda > 420$  nm. The time dependence of photo-generated currents was recorded by an electrochemical workstation (CHI 660e, CH Instrument, Shanghai, China). The photocurrents represent the photogeneration of H<sub>2</sub>O<sub>2</sub> by C<sub>3</sub>N<sub>4</sub> (Fig. S23a, black trace) and CDots-C<sub>3</sub>N<sub>4</sub> (Fig. S23b, black trace). The photocurrent of C<sub>3</sub>N<sub>4</sub>, rapidly decreases with time due to its poisoning by H<sub>2</sub>O<sub>2</sub> absorbed to its surface. The photocurrent of the CDots-C<sub>3</sub>N<sub>4</sub> catalyst is oscillating between 190-240 nA. We interpret these oscillations as cycles of H<sub>2</sub>O<sub>2</sub> generation, adsorption to the catalyst surface, decomposition by the CDots and diffusion out of the reaction region

We further used our scanning electrochemical microscope to study the diffusion of H<sub>2</sub>O<sub>2</sub>. Fig. S24a shows the H<sub>2</sub>O<sub>2</sub> oxidation current at the detector versus its X position during H<sub>2</sub>O<sub>2</sub> generation under light irradiation ( $\lambda > 420$  nm) indicating a diffusion length of  $\sim 200$   $\mu$ m. A Z scan of the current probe above a fixed x position (where the H<sub>2</sub>O<sub>2</sub> is maximal) shows again that the H<sub>2</sub>O<sub>2</sub> diffusion length is about 150  $\mu$ m. The limited diffusion length indicates that upon generation of H<sub>2</sub>O<sub>2</sub> on the C<sub>3</sub>N<sub>4</sub> surface its diffusion from its generation region is slow. The large concentration of H<sub>2</sub>O<sub>2</sub> in its generation region enhances its adsorption and increases the poisoning of the catalyst surface.

## 6. Electron-transfer number (n) calculation.

The electron-transfer number (n) is calculated as follows:

$$n = \frac{4I_{\text{disk}}}{I_{\text{disk}} + I_{\text{ring}}/N} \quad (10)$$

Here N is the RRDE collection efficiency, measured to be 0.24 using the reversible [Fe(CN)<sub>6</sub>]<sup>4-</sup>/[Fe(CN)<sub>6</sub>]<sup>3-</sup> system. In details, The RRDE electrode is dipped into 0.01 mol/L K<sub>3</sub>[Fe(CN)<sub>6</sub>] in 0.1 mol/L KCl solution and rotated under different rotation rates ( $\omega = 100, 400, 900, 1600$  rpm). The disk potential  $E_{\text{disk}}$  is scanned from 0.4 V to -0.6 V vs. SCE at a scan rate of 10 mV/s, the ring potential  $E_{\text{ring}}$  is fixed to 0.497 V vs. SCE (the reduced product [Fe(CN)<sub>6</sub>]<sup>4-</sup> can be oxidized at this potential), and the current-potential voltammograms are recorded during the electrode rotation. The RRDE measurement is shown in Fig. S25. The ratio of  $I_{\text{ring}}/I_{\text{disk}}$  is almost constant under various  $\omega$ . The averaged collection efficiency N is 0.24.

Based on the CV curve (Fig.S26) of the CDots-C<sub>3</sub>N<sub>4</sub> under dark conditions we have seen that, applying a positive voltage to the disk (0.4, 0.6 and 0.8 V), no electrochemical catalysis process for water oxidation is occurring. A pure photocatalytic water oxidation process was thus expected for a light irradiated CDots-C<sub>3</sub>N<sub>4</sub> solution under these voltage conditions. The electron transfer number was thus determined from the current of photo-induced water oxidation on the surface of CDots-C<sub>3</sub>N<sub>4</sub>, applying RRDE measurements.



As shown in Fig. S27, at the ring potential of 0.9 V, H<sub>2</sub>O<sub>2</sub> can be specific oxidized to O<sub>2</sub>. While, for ring potential at -0.3 V, other trace amount of oxygen species (e.g., O<sub>2</sub> and HO·) will be also reduced at this potential, which will affect the test results. So we selected 0.9 V (vs. SCE) as ring potential to detect H<sub>2</sub>O<sub>2</sub> generation. Table S3 shows the photoelectro-test results (electron transfer number) with ring potential at 0.9 V or -0.3 V (vs. SCE), and different positive voltage applied to the disk (0.4, 0.6 and 0.8 V). In all cases the electron transfer number was ~ 2.

## 7. Study of the 2e<sup>-</sup>/2e<sup>-</sup> two step mechanism of water splitting by the CDots-C<sub>3</sub>N<sub>4</sub> composite catalyst.

This section details a systematic study of the water splitting mechanism by the CDots-C<sub>3</sub>N<sub>4</sub> composite catalyst which employed both photocatalytic and photoelectrochemical experiments. Thermodynamically (Fig. 2D), C<sub>3</sub>N<sub>4</sub> is capable of producing H<sub>2</sub> and O<sub>2</sub> under visible light irradiation of water. Indeed, several groups have reported hydrogen evolution from water by C<sub>3</sub>N<sub>4</sub> with a sacrificial reductant (19); however the production of O<sub>2</sub> from water by C<sub>3</sub>N<sub>4</sub> is difficult (19, 29), on account of poisoning by intermediary H<sub>2</sub>O<sub>2</sub>, as noted earlier (19). We added *o*-tolidine as peroxide indicator to the reaction system containing pure water and photocatalysts, and monitored the production of H<sub>2</sub>O<sub>2</sub> by measuring the absorption peak of *o*-tolidine at 446 nm using UV-Vis spectroscopy. Note that we needed to strongly centrifuge the solution (12000 rpm) and to remove the catalyst to be able to suppress the UV absorption background of the catalyst. Fig. S28A shows that the absorption peak at 446 nm in the UV-Vis spectrum (black trace) was distinctly observed in the reaction system with C<sub>3</sub>N<sub>4</sub> as catalyst after irradiation for 12 h. The total H<sub>2</sub>O<sub>2</sub> production was measured to be 5.5 μmol. Meanwhile, control experiments using CDots as photocatalysts showed no H<sub>2</sub>O<sub>2</sub> in the reaction system (blue curve). Similarly, no H<sub>2</sub>O<sub>2</sub> was detected using a CDots-C<sub>3</sub>N<sub>4</sub> composite as catalyst (red trace in Fig. S28A). These results suggest that CDots play an important role in enabling the catalytic activity of CDots-C<sub>3</sub>N<sub>4</sub> by decomposing H<sub>2</sub>O<sub>2</sub> produced in the reaction.

The generation rate of H<sub>2</sub>O<sub>2</sub> from water splitting photocatalyzed by pure C<sub>3</sub>N<sub>4</sub> decreases with time (line slope of black blocks in Fig. S28B). The initial generation rate of 7.88 μmol/h (slope of Fig. S28B at t=0) is similar to the steady state generation rate of H<sub>2</sub> of ~8.5 μmol/h (Fig. S28B red dots) from water photocatalyzed by the non-optimized CDots-C<sub>3</sub>N<sub>4</sub> (CDots weight concentration 1.6x10<sup>-5</sup>). The H<sub>2</sub>O<sub>2</sub> generation rate catalyzed by pure C<sub>3</sub>N<sub>4</sub> however decreases with time due to poisoning of C<sub>3</sub>N<sub>4</sub> by H<sub>2</sub>O<sub>2</sub>. The above mentioned generation of 5.5 μmol H<sub>2</sub>O<sub>2</sub> in 12 hours reflects a much lower average generation rate than the initial one (by a factor of 17) due to poisoning. Additional photo-electrochemical measurements proving that the catalytic activity of C<sub>3</sub>N<sub>4</sub> is poisoned by the produced H<sub>2</sub>O<sub>2</sub> are presented in the supplementary material section 5. Next, experiments were performed to probe the role of CDots in catalyzing disproportionation of H<sub>2</sub>O<sub>2</sub>, which generates O<sub>2</sub> and eliminates C<sub>3</sub>N<sub>4</sub> poisoning. Pure CDots (0.67 mL, 2 mg/L) or 80 mg of CDots-C<sub>3</sub>N<sub>4</sub> composite (containing the same amount of 1.33 μg CDots) were introduced to 150 mL of 25 mM H<sub>2</sub>O<sub>2</sub> in ultrapure water. Fig. S28C (inset table) shows that the O<sub>2</sub> evolution rate under dark conditions for both the composite catalyst and the pure CDots was ~950 μmol/h, which is more than 100 times larger than the rate of the first step or H<sub>2</sub> evolution at ~8.5 μmol/h by CDots-C<sub>3</sub>N<sub>4</sub>. Fig. S28C also shows that light irradiation (λ>420 nm) only slightly (by ~12% for CDots-C<sub>3</sub>N<sub>4</sub> and by ~14% for CDots) increased the rate of the second stage or H<sub>2</sub>O<sub>2</sub> decomposition. These results prove that indeed CDots chemically catalyze H<sub>2</sub>O<sub>2</sub> decomposition (second step) at a rate two orders of magnitude larger than that of H<sub>2</sub> generation (first step). The first step or H<sub>2</sub>

generation is thus the rate-limiting step of the present water splitting process. Note that a small amount of CDots ( $\sim 1.6 \times 10^{-5}$  of the  $C_3N_4$  by weight) is sufficient to catalyze the decomposition of  $H_2O_2$  allowing a steady and efficient generation of both  $H_2$  and  $O_2$ . Additional electrochemical measurements proving the catalytic effect of the CDots are also presented in the supplementary material section 4.

Further experiments (performed using the non-optimized composite with a CDots concentration of  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) aimed to resolve the character of the water splitting process (either a four-electron one-step process or a  $2e^-/2e^-$  two-step process). To study the electron-transfer number, we applied rotating ring-disk electrode (RRDE) experiments (for full details see the supplementary material section 6) at a speed of 1600 rpm (Fig. S28D). The disk electrode was irradiated with visible light ( $\lambda > 420$  nm) and held at 0.6 V ( $E_{\text{disk}}$ ; vs. SCE). No electrochemical water oxidation was observed under these experimental conditions by independent cyclic voltammetry measurements (Fig. S26). The Pt ring was held at 0.9 V ( $E_{\text{ring}}$ ; vs. SCE). At this potential, the  $H_2O_2$  photogenerated on the disk could be oxidized to  $O_2$  on the ring. The electron transfer number can be calculated from equation (10) in the supplementary material section 6:

$$n = \frac{4I_{\text{disk}}}{I_{\text{disk}} + I_{\text{ring}}/N}$$

where N is the RRDE collection efficiency, measured to be 0.24. The electron-transfer number of the photo-induced water oxidation process is  $n \approx 2$  for CDots- $C_3N_4$  (Fig. S28D), the same as that for  $C_3N_4$  (Fig. S29), further confirming that water oxidation catalyzed by CDots- $C_3N_4$  proceeded via a two-electron process. The n values are constant for disk electrode potentials from 0.4 to 0.8 V and different light intensities (see Table S3). The above results confirm that water splitting photocatalysis by CDots- $C_3N_4$  indeed proceeds via the stepwise  $2e^-/2e^-$  two-step process, in which  $H_2O$  oxidation to  $H_2O_2$  is the first and rate-limiting step, followed by the second and fast step of  $H_2O_2$  disproportionation to  $O_2$  chemically catalyzed by CDots. We also studied the effect of the light intensity on the process, both in pure photocatalysis experiments (probing the  $H_2$  generation rate at different light intensities) and in RRDE electrochemical experiments (measuring ring and disk currents under different  $E_{\text{disk}}$  and  $E_{\text{ring}}$  voltages and light intensities to determine the electron transfer number (n), Fig. S28D, S29 and Table S3). The  $H_2$  generation rate (Fig. S30) and all measured currents increased linearly with the light intensity (Fig. S28D, S29 and Table S3). Moreover, under all light intensities  $n \approx 2$  (Fig. S28D, S29 and Table S3). The linear relationship between the light intensity and the gas generation rates/electrochemical currents indicate that no back reaction of any kind was present.

The above results confirm that water splitting photocatalysis by CDots- $C_3N_4$  indeed proceeds via the stepwise  $2e^-/2e^-$  two-step process, in which  $H_2O$  oxidation to  $H_2O_2$  is the first and rate-limiting step, followed by the second and fast step of  $H_2O_2$  disproportionation to  $O_2$  chemically catalyzed by CDots. Thus, the reaction mechanism shown in Fig. S32 is proposed to explain water splitting by the CDots- $C_3N_4$  photocatalyst. In brief, visible light illumination of a CDots- $C_3N_4$  catalyst generates conduction band electrons ( $e^-$ ) and valence band holes ( $h^+$ ). The photoexcited electrons in the conduction band of CDots- $C_3N_4$  are thermodynamically capable of water reduction to release  $H_2$ , while the holes in the valence band are capable of water oxidation to form  $H_2O_2$  via a  $2e^-$  process with a rate of  $\sim 8.5$   $\mu\text{mol/h}$  for a lower concentration of CDots in the catalyst ( $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>). When CDots- $C_3N_4$  with a higher concentration of CDots ( $4.8 \times 10^{-3}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) is used as the photocatalyst, the hydrogen evolution rate is increased to 46  $\mu\text{mol/h}$  (Fig. 4C). Consequently, in present system, the CDots- $C_3N_4$  photocatalyst can promote overall water splitting

via the stepwise  $2e^-/2e^-$  two-step process of  $H_2O_2$  formation and disproportionation with remarkably high efficiencies, although the concerted  $4e^-$  water splitting process is severely inhibited. The CDots- $C_3N_4$  catalyst achieves the aim of overall water splitting by visible light with unprecedented optimized efficiencies of 16% (for  $\lambda=420\pm20$  nm) without the need of any metal catalyst or sacrificial agent.

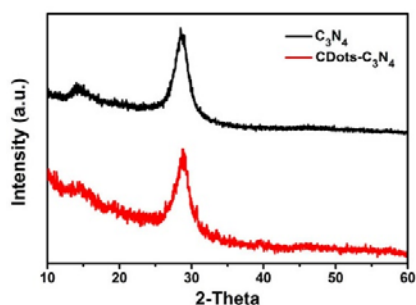
To end this section we note that complete water splitting *via* the two-step process is possible only for a CDots- $C_3N_4$  composite, but impossible for a powder mixture of CDots and  $C_3N_4$ . It is true that pure  $C_3N_4$  can photocatalyze water splitting to  $H_2$  and  $H_2O_2$  (first step), and pure CDots can chemically catalyze the decomposition of  $H_2O_2$  (second step). Pure CDots are capable of decomposing  $H_2O_2$  in a solution, but not when  $H_2O_2$  is adsorbed to the  $C_3N_4$  surface. In order for the two water splitting steps to occur successively and synergistically (i.e. to enable disproportionation of adsorbed  $H_2O_2$ ) the CDots must be distributed in the closest proximity to the active surface of the porous  $C_3N_4$  grains, which is achieved in the form of a CDots- $C_3N_4$  composite.

#### **8. The stability of the CDots- $C_3N_4$ catalyst under low load (10 mg/150 mL).**

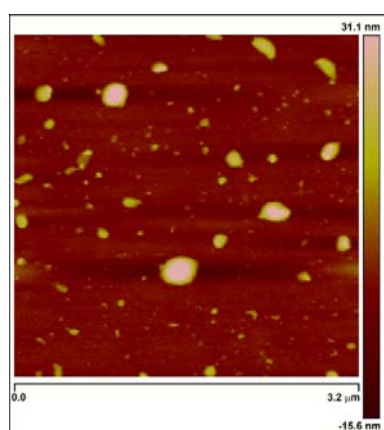
Most of our gas generation stability and mass loss experiments were conducted for a high load of the catalyst of 80 mg/150 mL. Such a high load may compensate for catalyst losses (serving as a reservoir) that might be therefore better observed for a low load of the catalyst. We therefore used a lower density catalyst solution (10 mg in 150 mL water) to evaluate its stability. A time course of  $H_2$  production from water under visible light ( $\lambda>420$  nm) (applying a catalyst with CDots concentration of  $4.8\times10^{-3}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) is shown in Fig. S15. The linear relation between  $H_2$  production and the reaction time indicates that the photocatalytic activity of the catalyst was stable for the entire experiments (half a month). The mass change of the CDots- $C_3N_4$  ( $4.8\times10^{-3}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) applied using the low density solution (10 mg in 150 mL water) was detected as well. The catalyst was collected and dried in a vacuum oven at 60 °C after each 24 h cycle of usage. The mass of the recycled powder was weighted by the electronic balance before and after light irradiation is listed in Table S2. No apparent mass loss could be observed within the experimental error ( $\pm0.0008$  g).

#### **9. Additional information.**

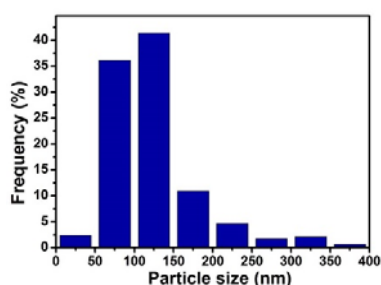
Digital photos of catalyst solution and the setup we used for water splitting experiment are shown in Fig. S33 and S34. XPS survey spectrum and FT-IR spectrum of CDots are listed in Fig. S35. The additional TEM characterizations performed show that the SAED of the pure  $C_3N_4$  matrix (Fig. S36a) exhibits crystalline lobes while those of the composite material (Fig. S36b) are hazy. This indicates that the matrix of the  $4.8\times10^{-3}$  CDots- $C_3N_4$  composite is more disordered with respect to the pure matrix.



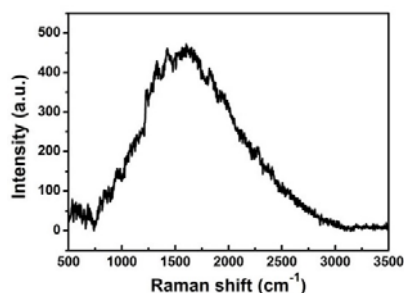
**Fig. S1.** Experimental XRD patterns for carbon nitride ( $C_3N_4$  and  $CDots-C_3N_4$ ), revealing a graphitic structure with an interplanar stacking distance of aromatic units of 0.326 nm. The sharp  $\langle 002 \rangle$  diffraction peak at  $27.9^\circ$  indicates the characteristic interlayer stacking structure of aromatic systems, revealing an interlayer spacing of about 0.324 nm (38,44). While the  $\langle 100 \rangle$  diffraction peak at  $13.8^\circ$  relates to the interplanar structural packing, such as the hole-to-hole distance of the nitride pores in the crystal, corresponding to an interlayer spacing of 0.675 nm, which is similar to the XRD patterns of  $C_3N_4$  in the previous works (45). The average size of  $C_3N_4$  and  $CDots-C_3N_4$  are  $\sim 4$  nm from the line width analysis of the  $\langle 002 \rangle$  diffraction peak using Scherrer' equation.



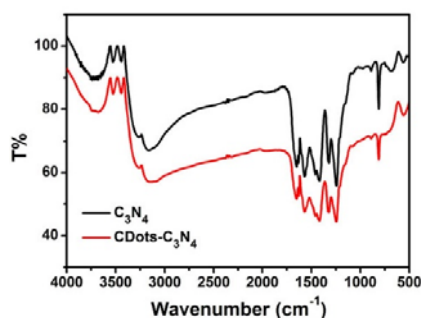
**Fig. S2** The tapping-mode atomic force microscopy (AFM) image of the as-synthesized  $CDots-C_3N_4$  composite grains.



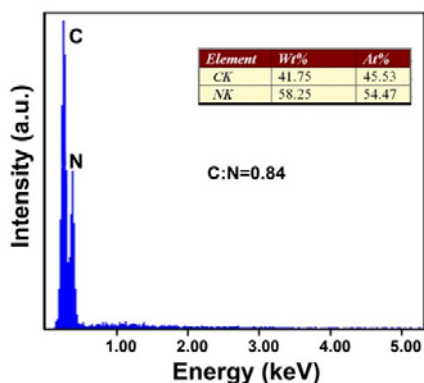
**Fig. S3** Size distribution of the grains of the as-prepared  $CDots-C_3N_4$  composites (from the AFM image, Fig. S2).



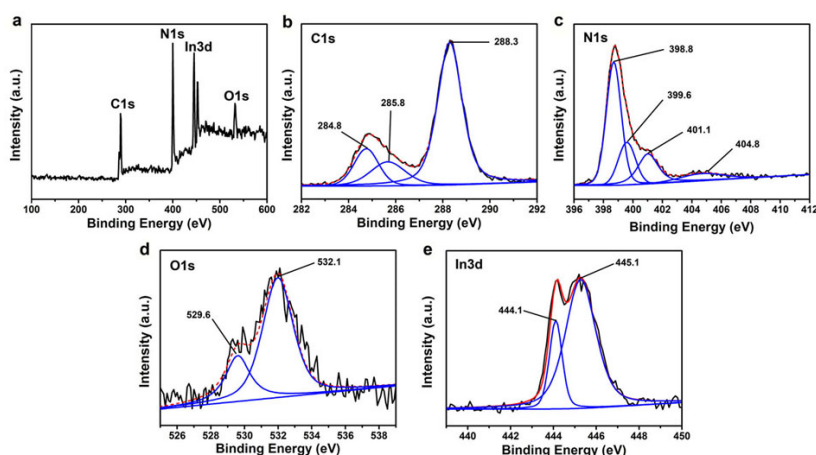
**Fig. S4** Raman spectrum of CDots-C<sub>3</sub>N<sub>4</sub> (CDots concentration of  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) excited by 633 nm laser. The broad symmetric peak in the range 700-3000 cm<sup>-1</sup> is attributed to the overlap of several peaks: N=C=N at 1400 cm<sup>-1</sup>, -C≡N at 1600 cm<sup>-1</sup>, and atmospheric stretching vibration at 2300 cm<sup>-1</sup>(46). Moreover, this Raman spectrum is similar to that of the pure C<sub>3</sub>N<sub>4</sub>, suggesting that the structure of C<sub>3</sub>N<sub>4</sub> is well preserved in CDots-C<sub>3</sub>N<sub>4</sub> after introduction of CDots.



**Fig. S5** FT-IR spectra of the as-obtained graphitic carbon nitride powders (C<sub>3</sub>N<sub>4</sub>-black and CDots-C<sub>3</sub>N<sub>4</sub>-red. CDots concentration of  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>). The C<sub>3</sub>N<sub>4</sub> spectrum contains several major bands between 1200 cm<sup>-1</sup> and 1650 cm<sup>-1</sup>, which correspond to the characteristic stretching modes of CN heterocycles. In addition, the breathing mode of triazine units appears at 809 cm<sup>-1</sup>. The results are consistent with former reports.<sup>1</sup> The characteristic features of the condensed C-N heterocycles, i.e., the typical breathing mode at 806.5 cm<sup>-1</sup> and stretching mode in 1200–1650 cm<sup>-1</sup> of the tri-s-triazine units are also found in the IR spectra of CDots-C<sub>3</sub>N<sub>4</sub> (red curve). Specifically, the bands at 1315.7 and 1241.2 cm<sup>-1</sup> respectively correspond to the stretching vibrations of the connected units of N-(C)<sub>3</sub> (full condensation) and C-N-C (partial condensation), while the bands at 1645.4, 1564.5, and 1408.3 cm<sup>-1</sup> represent the stretching vibration modes of heptazine-derived repeating units (47).

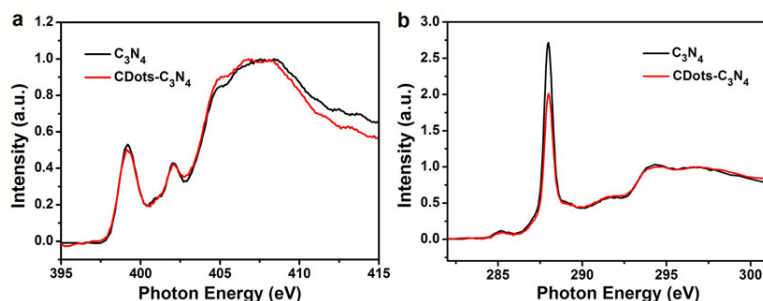


**Fig. S6** EDS spectra of CDots-C<sub>3</sub>N<sub>4</sub> (CDots concentration of  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) prepared by pyrolysis of urea at 550 °C. The Energy Dispersive Spectroscopy (EDS) spectrum clearly shows that carbon (C) and nitrogen (N) are the main constituent elements. The atomic N: C ratio for the as-produced CDots-C<sub>3</sub>N<sub>4</sub> is 1.19, which is lower than the theoretical atomic N:C ratio (1.33) of C<sub>3</sub>N<sub>4</sub> (48), suggesting that CDots have been incorporated into the C<sub>3</sub>N<sub>4</sub> matrix.

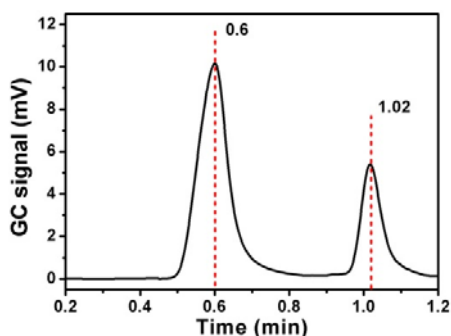


**Fig. S7** XPS spectra of CDots-C<sub>3</sub>N<sub>4</sub> prepared by pyrolysis of urea at 550 °C (CDots concentration  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>). The CDots-C<sub>3</sub>N<sub>4</sub> powder was placed on an In substrate, which served as a calibration material for determination of the correct binding energy. Most of the O1S peak signal comes from the In substrate. (a) XPS survey spectrum. (b) C1s spectrum. (c) N1s spectrum. (d) O1s spectrum. (e) In3d spectrum. The XPS survey spectrum in Fig. S7a shows that CDots-C<sub>3</sub>N<sub>4</sub> is primarily composed of carbon and nitrogen. The C1s spectrum (Fig. S7b) shows three deconvoluted peaks at 288.3 eV, 285.8 eV and 284.8 eV. The major C peak at 288.3 eV is identified as sp<sup>2</sup>-bonded carbon (N–C=N), and the weaker one at 284.8 eV corresponds to graphitic carbon, which is usually observed in the XPS spectrum of carbon nitrides. As shown in Fig. S7c, N1s peak at 398.8 eV corresponds to sp<sup>2</sup> hybridized aromatic N bonded to carbon atoms (C=N–C). The peak at 399.6 eV is assigned to the tertiary N bonded to carbon atoms in the form of N–(C)<sub>3</sub> or H–N–(C)<sub>2</sub>. The weaker peak with a high binding energy at 401.1 eV is attributed to quaternary N bonded to three carbon atoms in the aromatic cycles (49–50). And the peak centered at 404.8 eV corresponds to the  $\pi$  excitations (51–52). The O1s core level spectrum (Fig. S7d) has

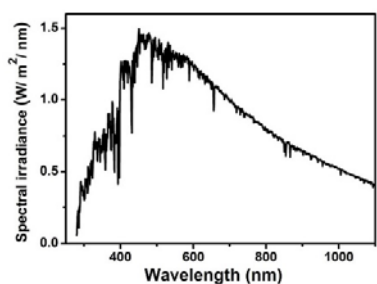
two distinct components. The O 1s peaks at 529.6 and 532.1 eV from oxygen atoms are attributed to Indium oxide and adsorbed water, respectively. In Fig. S7e, the lower binding energy peak is due to the Indium (Indium oxide) substrate.



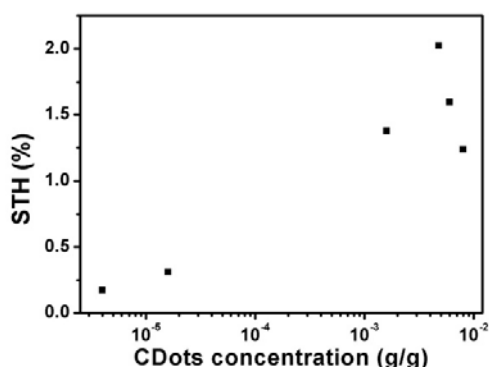
**Fig. S8** X-ray absorption near edge structure (XANES) spectroscopy of  $C_3N_4$  (black trace) and CDots- $C_3N_4$  (red trace) at the N 1s line (a) and at the C 1s line (b). The CDots concentration is  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>. The nitrogen K-edge XANES spectra (Fig. S8a) of CDots- $C_3N_4$  and  $C_3N_4$  are almost identical, while the carbon K-edge XANES spectra of CDots- $C_3N_4$  (Fig. S7b, red trace) and  $C_3N_4$  (Fig. S8b, black trace) are also very similar, where the first sharp peak at 288 eV corresponds to  $\pi^*$  bond formed by the C-N ring. The decrease of the peak intensity of the CDots- $C_3N_4$  at 288 eV represents a corresponding reduction in the number of  $\pi^*$  bonds, which indicates that CDots introduction may locally affect the  $C_3N_4$  matrix.



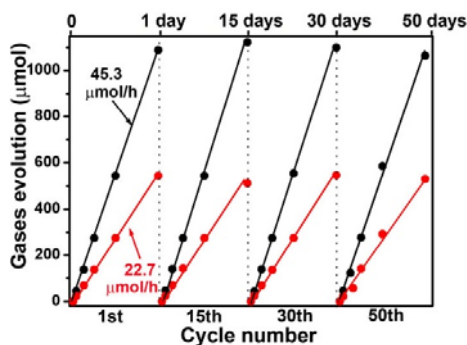
**Fig. S9** A typical gas chromatograph trace of evolved hydrogen and oxygen. The gas chromatograph signal peaks at 0.6 and 1.02 min are attributed to  $H_2$  and  $O_2$  evolution respectively.



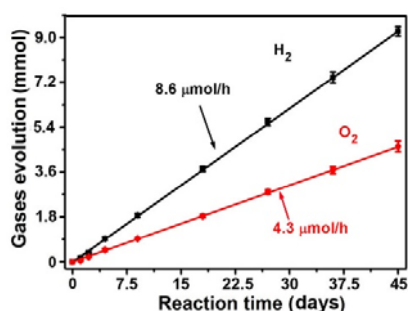
**Fig. S10** Typical output spectral distribution of AM 1.5G solar simulator.



**Fig. S11** Solar-to-hydrogen efficiency (STH) for different concentrations of CDots ( $\text{g}_{\text{CDots}}/\text{g}_{\text{catalyst}}$ ) in a fixed mass of composite catalyst. The experimental conditions were: 0.080 g catalyst, 150 mL ultrapure water, solar simulator irradiation for 6 h.

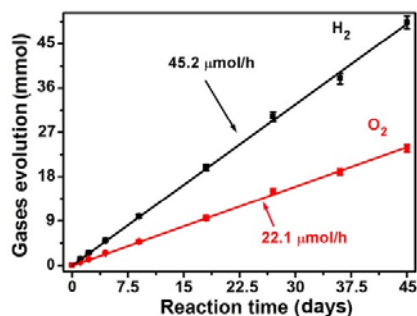


**Fig. S12** A time course of H<sub>2</sub> and O<sub>2</sub> production from water under visible light ( $\lambda > 420 \text{ nm}$ ) by CDots-C<sub>3</sub>N<sub>4</sub> ( $4.8 \times 10^{-3} \text{ g}_{\text{CDots}}/\text{g}_{\text{catalyst}}$ ). After each 1-day usage the catalyst was dried and reused for the next cycle. Note the steady production rate of H<sub>2</sub> and O<sub>2</sub> over extended periods. No gases (CO<sub>2</sub>, N<sub>2</sub> etc.) other than H<sub>2</sub> and O<sub>2</sub> were released.

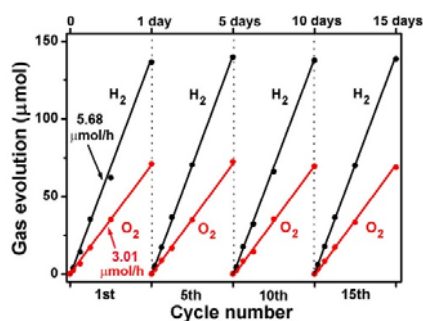


**Fig. S13** The 45-days time course of H<sub>2</sub> and O<sub>2</sub> evolution from water photo-catalyzed by CDots-C<sub>3</sub>N<sub>4</sub> ( $1.6 \times 10^{-5} \text{ g}_{\text{CDots}}/\text{g}_{\text{catalyst}}$ ) composites under visible light ( $\lambda > 420 \text{ nm}$ ) irradiation. Note the linear evolution with time indicating steady and stable water splitting. The error bars indicate the maximum and minimum values obtained, while the dot represents the average value.

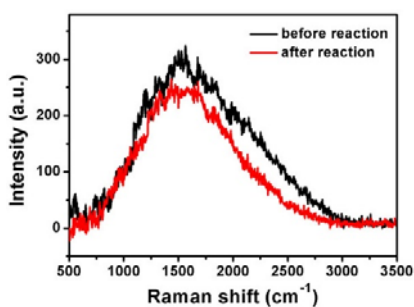




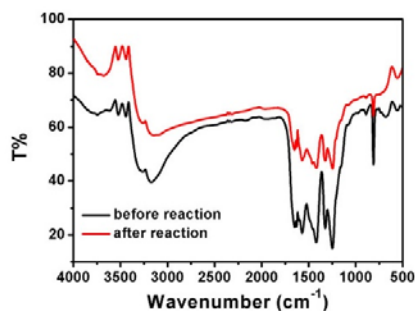
**Fig. S14** The 45-days time course of H<sub>2</sub> and O<sub>2</sub> evolution from water photo-catalyzed by CDots-C<sub>3</sub>N<sub>4</sub> ( $4.8 \times 10^{-3}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) composites under visible light ( $\lambda > 420$  nm) irradiation. Note the linear evolution with time indicating stable and steady water splitting. The error bars indicate the maximum and minimum values obtained, while the dot represents the average value.



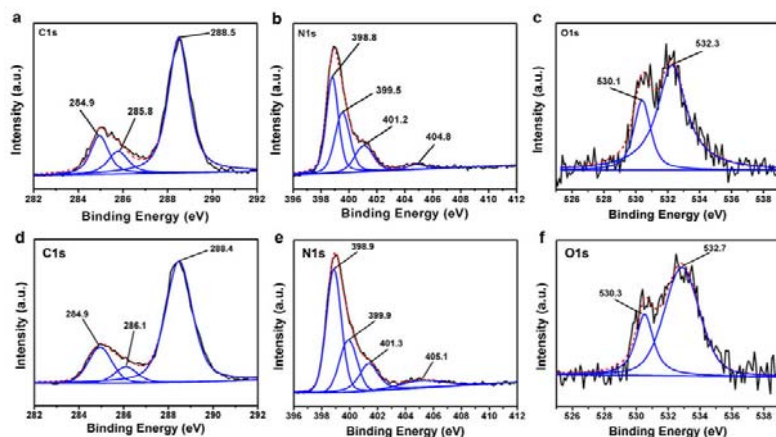
**Fig. S15** A time course of H<sub>2</sub> and O<sub>2</sub> production from water under visible light ( $\lambda > 420$  nm) by 10 mg CDots-C<sub>3</sub>N<sub>4</sub> ( $4.8 \times 10^{-3}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) in 150 mL (low loading of catalyst). Note the stable production rate of H<sub>2</sub> and O<sub>2</sub> over 15 cycles.



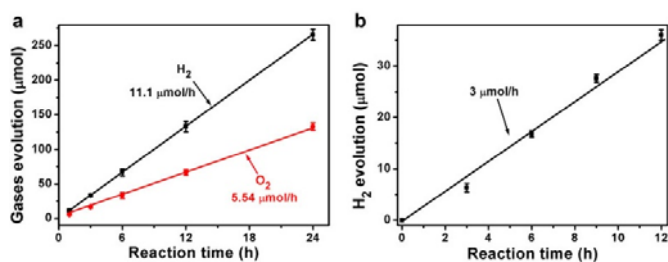
**Fig. S16** Raman spectrum of CDots-C<sub>3</sub>N<sub>4</sub> (CDots concentration of  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) excited by 633 nm laser before and after 50 cycles of 24h water splitting.



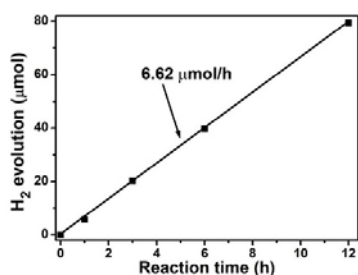
**Fig. S17** FT-IR spectra of CDots-C<sub>3</sub>N<sub>4</sub> powders (CDots concentration of  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) before and after 50 cycles of 24 h water splitting.



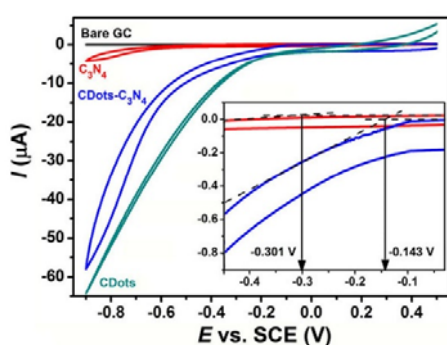
**Fig. S18** XPS spectra of CDots-C<sub>3</sub>N<sub>4</sub> (CDots concentration  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) before (a-c) and after (d-f) 50 cycles of 24 h water splitting. (a, d) C1s spectrum. (b, e) N1s spectrum. (c, f) O1s spectrum.



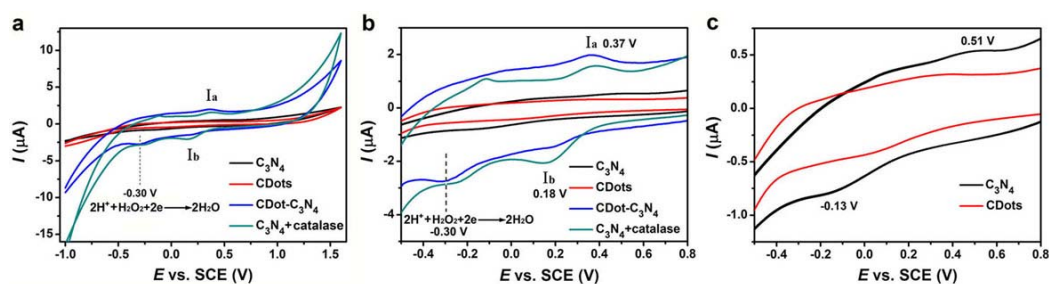
**Fig. S19** (a) The photocatalytic activity of CDots-C<sub>3</sub>N<sub>4</sub> composites ( $4.8 \times 10^{-3}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) in sea water. The time course of H<sub>2</sub> and O<sub>2</sub> production from sea water under visible light irradiation by a 300 W Xenon lamp using a cut-off filter ( $\lambda > 420$  nm) yields an evolution rate of 11.1  $\mu\text{mol/h}$ , 24% of the evolution rate from ultra-pure water (46  $\mu\text{mol/h}$ ) under the same conditions. (b) The photocatalytic activity of CDots-C<sub>3</sub>N<sub>4</sub> composites for QE determination. A time course of H<sub>2</sub> production from sea water under 300 W Xenon lamp using a  $420 \pm 20$  nm band-pass filter by CDots-C<sub>3</sub>N<sub>4</sub> ( $4.8 \times 10^{-3}$  g<sub>CDots</sub>/g<sub>catalyst</sub>). The linear evolution with time indicating steady water splitting with time. The error bars indicate the maximum and minimum values obtained, while the dot represents the average value.



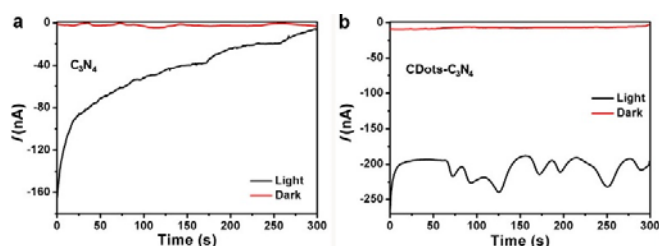
**Fig. S20** The photocatalytic activity of CDots-C<sub>3</sub>N<sub>4</sub> composites for QE determination. A time course of H<sub>2</sub> production from water under 300 W Xenon lamp using a 420±20 nm band-pass filter by CDots-C<sub>3</sub>N<sub>4</sub> ( $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>). The linear evolution with time indicating steady water splitting with time.



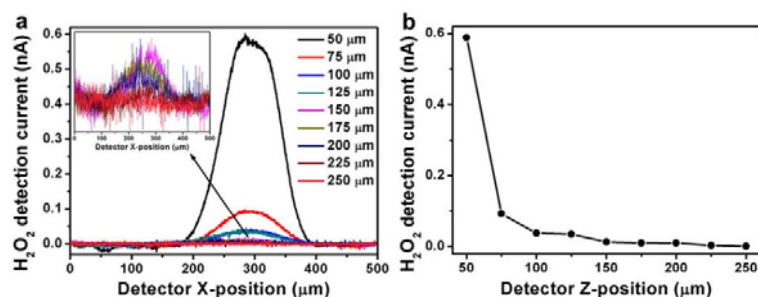
**Fig. S21** Cyclic voltammograms of bare GC-(black trace), C<sub>3</sub>N<sub>4</sub>-(red trace), CDots-(green trace), and CDots-C<sub>3</sub>N<sub>4</sub>-(blue trace, CDots concentration of  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) modified GC electrodes in 0.05 M (pH=7.0) phosphate buffered 25 mM H<sub>2</sub>O<sub>2</sub> solution at a scan rate of 50 mV/s (without light).



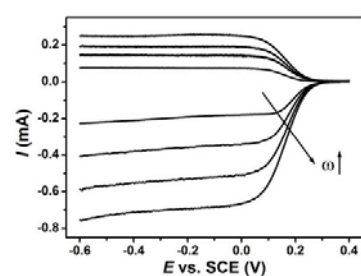
**Fig. S22** (a) Cyclic voltammograms of CDots (black trace), C<sub>3</sub>N<sub>4</sub> (red trace), C<sub>3</sub>N<sub>4</sub>-catalase (green trace), and CDots-C<sub>3</sub>N<sub>4</sub> (blue trace, CDots concentration of  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) in N<sub>2</sub>-saturated ultrapure water with light irradiation of  $\lambda > 420$  nm (scan rate: 50 mV s<sup>-1</sup>). (b) Partial cyclic voltammograms of CDots-C<sub>3</sub>N<sub>4</sub> (blue trace) and C<sub>3</sub>N<sub>4</sub>-catalase (green trace). (c) Partial cyclic voltammograms of C<sub>3</sub>N<sub>4</sub> (black trace) and CDots (red trace). The experiments were carried out in N<sub>2</sub>-saturated ultrapure water with light irradiation of  $\lambda > 420$  nm (scan rate: 50 mV s<sup>-1</sup>).



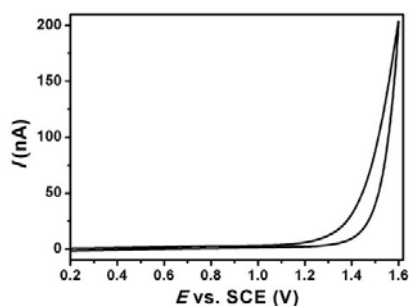
**Fig. S23** The current-time curve of (a)  $C_3N_4$  and (b)  $CDots-C_3N_4$  ( $4.8 \times 10^{-3}$  g $_{CDots}$ /g $_{catalyst}$ ). The photocurrents were measured at a bias which nulls the current under dark conditions in ultrapure water solution under 300 mW  $cm^{-2}$  Xenon lamp with a  $\lambda > 420$  nm cut-off filter. The  $C_3N_4$  curve (black trace in a) shows a rapid decrease of the photocurrent indicating poisoning of the catalyst surface by  $H_2O_2$ . The  $CDots-C_3N_4$  curve (black trace in b) shows current oscillations between 190-240 nA due to cycles of  $H_2O_2$  generation, adsorption, decomposition by CDots and diffusion.



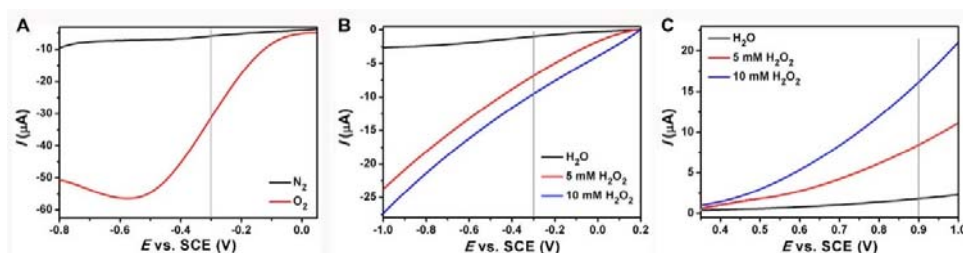
**Fig. S24** SECM detection of  $H_2O_2$  concentration (roughly) at different distances from the  $C_3N_4$  catalyst in the X and the Z directions. (a) The tip current as a function of the displacement of the tip electrode from the starting position in the x direction for different Z positions indicated in the legend. The detection was performed using ultrapure water with the tip potential held at 0.9 V vs. SCE. No voltage was applied on the FTO generator. (b)  $H_2O_2$  detection current during  $H_2O$  oxidation at the FTO electrode used as a generator (diameter: 1 mm, electrode potential: 0.9 V vs. SCE) versus the generator-detector distance. The diffusion length in both X and Z directions is smaller than 200  $\mu m$ .



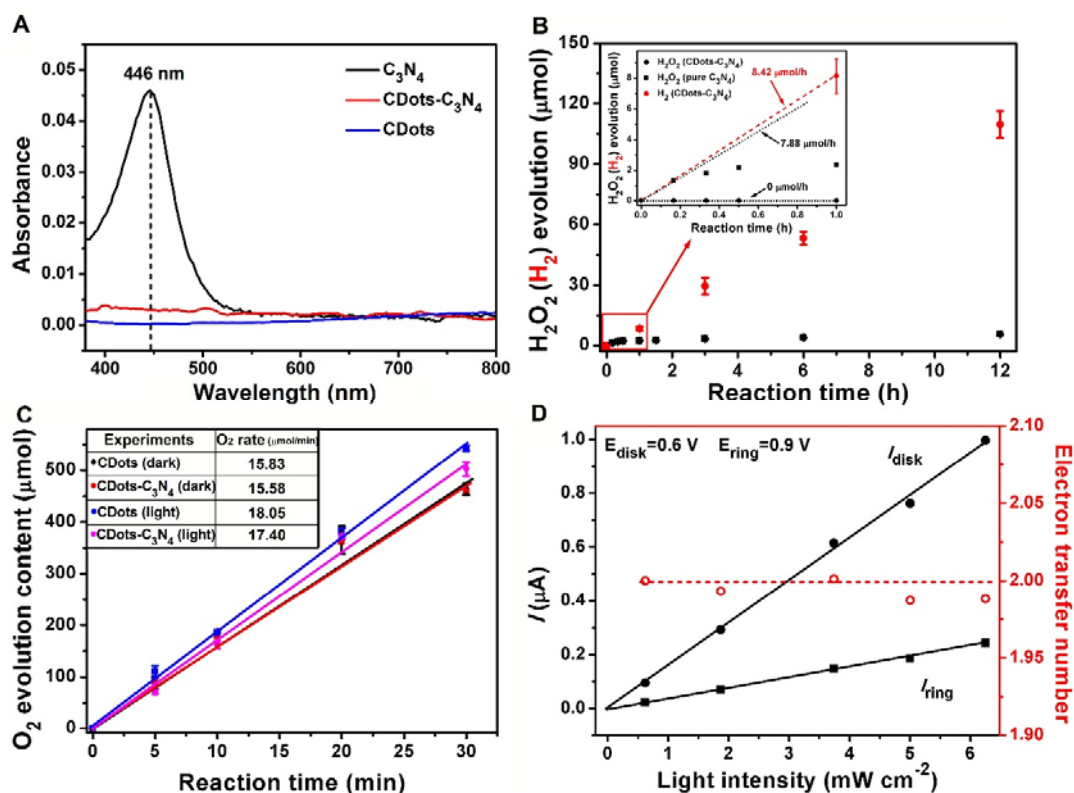
**Fig. S25** I-V curves of bare RRDE in 0.1 M KCl containing 0.01 M  $K_3Fe(CN)_6$  under rotation rates of: 100, 400, 900 and 1600 rpm (from inner to outer). These curves were used to determine the collection efficiency  $N$ .



**Fig.S26** CV curve collected by the tip biased at -0.3 V (vs. SCE) of an FTO electrode modified with CDots-C<sub>3</sub>N<sub>4</sub> (CDots concentration of  $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) in phosphate buffer at pH=7.0 under dark. No electrochemical oxidation reactions occur at 0.4-0.8 V (vs. SCE).

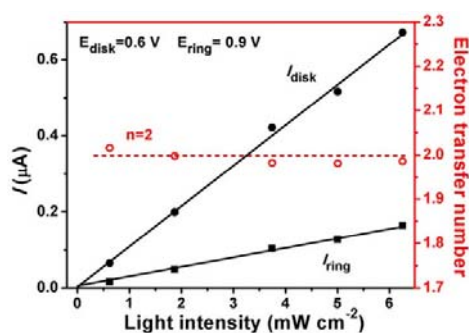


**Fig. S27** (A) Linear sweep voltammograms (LSV) curves of Pt ring electrode scan from 0.05 to -0.8 V (vs. SCE) in O<sub>2</sub>- and N<sub>2</sub>-saturated ultrapure water solutions at a 10 mV s<sup>-1</sup> scanning speed. (B) LSV curves of Pt ring electrode scan from 0.2 to -1.0 V (vs. SCE) in N<sub>2</sub>-saturated ultrapure water solution (black trace), 5 mM H<sub>2</sub>O<sub>2</sub> solution (red trace) and 10 mM H<sub>2</sub>O<sub>2</sub> solution (blue trace) at a 10 mV s<sup>-1</sup> scanning speed. (C) LSV curves of Pt ring electrode scan from 0.3 to 1.0 V (vs. SCE) in N<sub>2</sub>-saturated ultrapure water solution (black trace), 5 mM H<sub>2</sub>O<sub>2</sub> solution (red trace) and 10 mM H<sub>2</sub>O<sub>2</sub> solution (blue trace) at a 10 mV s<sup>-1</sup> scanning speed.

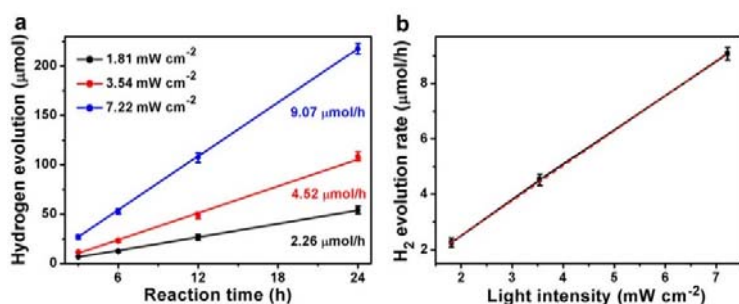


**Fig.S28** (A) Free  $H_2O_2$  in the reaction solution after 12 h light irradiation of  $\lambda > 420$  nm. The reaction system consists of 80 mg catalyst dispersed in 150 mL water using  $C_3N_4$  (black trace),  $CDots$  (blue trace), or  $CDots-C_3N_4$  (red trace) as catalyst. (B) A time course of  $H_2O_2$  production from water under visible light ( $\lambda > 420$  nm) by pure  $C_3N_4$  (black blocks) and  $H_2$  production by  $CDots-C_3N_4$ , with a  $CDots$  concentration of  $1.6 \times 10^{-5}$  g $_{CDots}$ /g $_{catalyst}$  (red dots) for 12 h. The initial  $H_2O_2$  production rate (7.88  $\mu mol/h$ , black blocks) is similar to the average generation rate of  $H_2$  (8.42  $\mu mol/h$ , red dots) as also marked by the corresponding tangent lines in the inset. The  $H_2O_2$  production from water by  $CDots-C_3N_4$  is zero (black dots in the inset pattern of Fig. S28B). The detection of  $H_2O_2$  is measured by *o*-tolidine. The amount of  $H_2O_2$  is calculated by the corresponding calibration curve (Fig. S31). (C)  $H_2O_2$  decomposition over  $CDots-C_3N_4$  versus time using pure  $C_3N_4$  or  $CDots-C_3N_4$  as catalysts under either dark or light irradiation ( $\lambda > 420$  nm). Note that pure  $CDots$  and  $CDots-C_3N_4$  catalyze the same  $H_2O_2$  decomposition rates and light irradiation has a minor effect on the decomposition rates ( $\sim 12\%$ - $14\%$  more than that in dark). (D) Pt disk-Pt ring electrodes (RRDE) measurements applying a rotation rate of 1600 rpm. The disk electrode is under  $\lambda > 420$  nm irradiation (with different light intensities) with potential kept at 0.6 V (vs. SCE). The Pt ring was biased at 0.9 V vs. SCE. Note the linear increase of the photocurrents with light intensity indicating no back reactions occur. Also note that the same charge transfer number of 2 (pointing to a  $2e^-$  process) was obtained for all light intensities. Non-optimized  $CDots-C_3N_4$  catalyst ( $CDots$  concentration,  $1.6 \times 10^{-5}$  g $_{CDots}$ /g $_{catalyst}$ ) was used in the experiments of Fig. S28. For S28B and C, the error bars indicate the maximum and minimum values obtained, while the dot represents the average value.

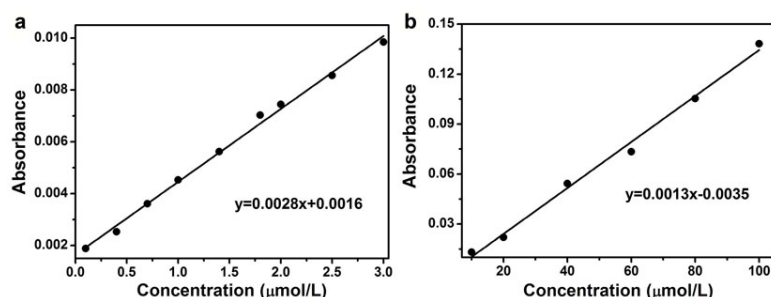




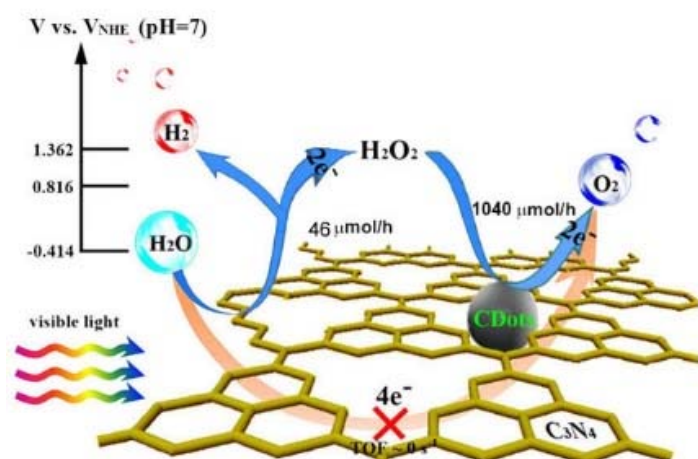
**Fig. S29** Pt disk-Pt ring electrodes (RRDE) over  $C_3N_4$  applying a rotation rate of 1600 rpm. The disk electrode is  $\lambda > 420$  nm irradiation (with different light intensity) with potential kept at 0.6 V (vs. SCE). The Pt ring was biased at 0.9 V vs. SCE. Note the linear depends of the currents on light intensity and the constant  $n$  obtained for all light intensities.



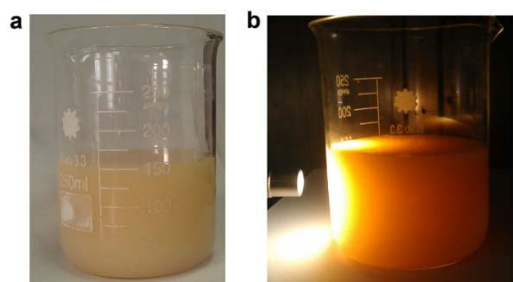
**Fig. S30** (a) Hydrogen evolution as a function of reaction time for CDots- $C_3N_4$  ( $4.8 \times 10^{-3}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) under irradiation by different light power densities. (b) The  $H_2$  evolution rate as a function of the light power density calculated from Fig. S30a. Note the linear relationship between the evolution rate and the light power density indicating no back reactions exist. The error bars indicate the maximum and minimum values obtained, while the dot represents the average value.



**Fig. S31** The calibration curve and the fitting equation of the  $H_2O_2$  concentration and absorbance, which are determined by external standard method using  $H_2O_2$  as standard materials with different concentration ((a) concentration is from 0.1 to 3.0  $\mu\text{mol/L}$ ; (b) from 10 to 100  $\mu\text{mol/L}$ ) under the same experimental conditions.



**Fig. S32** The proposed reaction mechanism for visible-light-driven water splitting by CDots-C<sub>3</sub>N<sub>4</sub>.

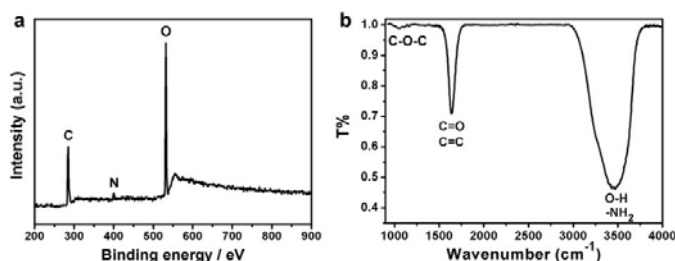


**Fig. S33** Digital photos of 80 mg CDots-C<sub>3</sub>N<sub>4</sub> dispersed in 150 mL ultrapure water. (a) Indoors at 10:00 a.m.. (b) With 300 W Xenon lamp irradiation from the left side.

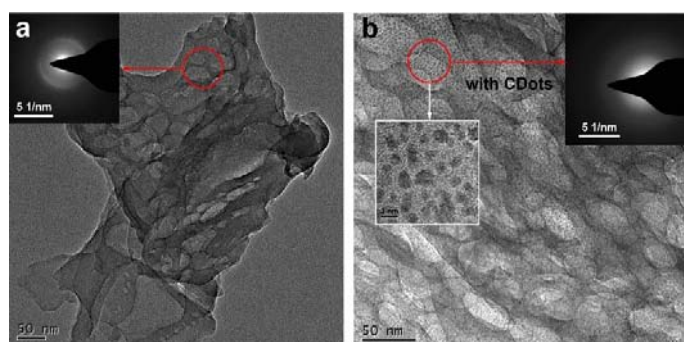


**Fig. S34** Photocatalysis equipment picture of our water splitting system.





**Fig. S35** (a) XPS survey and (b) FT-IR spectrum of CDots after treated by ammonia in Teflon stainless steel autoclave at 150 °C for 7 h (experimental details see preparation of CDots section in supplementary information). The N/C ratio of the CDots derived from XPS analysis (Fig. S35a) is about 0.08. The high O content should come from surface oxygen-containing groups (hydroxyl and carboxyl), which is further confirmed by FT-IR spectrum (Fig. S35b). In Fig. S35b, a broad peak at 3300-3500 cm<sup>-1</sup> suggests the existence -NH<sub>2</sub> and -OH groups (53). The peaks around 1100 cm<sup>-1</sup> denoted the stretching vibrations of C-O-C bonds, while the band at 1636 cm<sup>-1</sup> represents the C=O stretching vibration band (54).



**Fig. S36** TEM images of (a) C<sub>3</sub>N<sub>4</sub> and (b) CDots-C<sub>3</sub>N<sub>4</sub> (CDots concentration of  $4.8 \times 10^{-3}$  g<sub>CDots</sub>/g<sub>catalyst</sub>). The insets images show the SAED pattern of the selected area of TEM image marked with red circle. The SAED of the pure C<sub>3</sub>N<sub>4</sub> matrix (Fig. S36a) exhibits crystalline lobes while those of the composite material (Fig. S36b) are hazy. This indicates that the matrix of the  $4.8 \times 10^{-3}$  CDots-C<sub>3</sub>N<sub>4</sub> composite is more disordered with respect to the pure matrix.

**Table S1** CDots-C<sub>3</sub>N<sub>4</sub> mass following 24 h cycles of water splitting with high catalyst loading (80mg/150mL).

|                  | CDots-C <sub>3</sub> N <sub>4</sub> ( $1.6 \times 10^{-5}$ g <sub>CDots</sub> /g <sub>catalyst</sub> )/g | CDots-C <sub>3</sub> N <sub>4</sub> ( $4.8 \times 10^{-3}$ g <sub>CDots</sub> /g <sub>catalyst</sub> )/g |
|------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Initial mass     | 0.0804                                                                                                   | 0.0807                                                                                                   |
| After 1 cycle    | 0.0806                                                                                                   | 0.0801                                                                                                   |
| After 10 cycles  | 0.0801                                                                                                   | 0.0802                                                                                                   |
| After 30 cycles  | 0.0802                                                                                                   | 0.0806                                                                                                   |
| After 50 cycles  | 0.0796                                                                                                   | 0.0798                                                                                                   |
| After 200 cycles | 0.0784                                                                                                   | -----                                                                                                    |

**Table S2** CDots-C<sub>3</sub>N<sub>4</sub> ( $4.8 \times 10^{-3}$  g<sub>CDots</sub>/g<sub>catalyst</sub>) mass following 24h cycles of water splitting with low catalyst loading (10mg/150mL).

|                 | CDots-C <sub>3</sub> N <sub>4</sub> ( $4.8 \times 10^{-3}$ g <sub>CDots</sub> /g <sub>catalyst</sub> )/g |
|-----------------|----------------------------------------------------------------------------------------------------------|
| Initial mass    | 0.0099                                                                                                   |
| After 1 cycle   | 0.0107                                                                                                   |
| After 3 cycles  | 0.0102                                                                                                   |
| After 5 cycles  | 0.0097                                                                                                   |
| After 10 cycles | 0.0099                                                                                                   |
| After 15 cycles | 0.0106                                                                                                   |

**Table S3** Transfer number determined under different photo-electro-chemical test conditions. CDots-C<sub>3</sub>N<sub>4</sub>: ( $1.6 \times 10^{-5}$  g<sub>CDots</sub>/g<sub>catalyst</sub>).

|                                                                      | $E_{\text{disk}}$<br>Light intensity | 0.4 V | 0.6 V | 0.8 V |
|----------------------------------------------------------------------|--------------------------------------|-------|-------|-------|
| C <sub>3</sub> N <sub>4</sub><br>( $E_{\text{ring}} = -0.3$ V)       | 0.62 mW cm <sup>-2</sup>             | 2.025 | 1.972 | 1.972 |
|                                                                      | 1.87 mW cm <sup>-2</sup>             | 1.980 | 2.040 | 2.016 |
|                                                                      | 3.74 mW cm <sup>-2</sup>             | 2.032 | 2.024 | 2.011 |
|                                                                      | 5.0 mW cm <sup>-2</sup>              | 2.041 | 2.009 | 1.964 |
|                                                                      | 6.25 mW cm <sup>-2</sup>             | 1.971 | 1.989 | 1.978 |
| C <sub>3</sub> N <sub>4</sub><br>( $E_{\text{ring}} = 0.9$ V)        | 0.62 mW cm <sup>-2</sup>             | 1.976 | 2.016 | 2.019 |
|                                                                      | 1.87 mW cm <sup>-2</sup>             | 1.996 | 1.997 | 1.984 |
|                                                                      | 3.74 mW cm <sup>-2</sup>             | 2.014 | 1.982 | 1.993 |
|                                                                      | 5.0 mW cm <sup>-2</sup>              | 2.041 | 1.981 | 2.026 |
|                                                                      | 6.25 mW cm <sup>-2</sup>             | 1.982 | 1.986 | 2.036 |
| CDots-C <sub>3</sub> N <sub>4</sub><br>( $E_{\text{ring}} = -0.3$ V) | 0.62 mW cm <sup>-2</sup>             | 2.000 | 1.998 | 1.998 |
|                                                                      | 1.87 mW cm <sup>-2</sup>             | 1.997 | 2.009 | 2.000 |
|                                                                      | 3.74 mW cm <sup>-2</sup>             | 2.000 | 2.001 | 2.000 |
|                                                                      | 5.0 mW cm <sup>-2</sup>              | 1.999 | 2.005 | 1.999 |
|                                                                      | 6.25 mW cm <sup>-2</sup>             | 2.000 | 1.995 | 1.999 |
| CDots-C <sub>3</sub> N <sub>4</sub><br>( $E_{\text{ring}} = 0.9$ V)  | 0.62 mW cm <sup>-2</sup>             | 1.990 | 2.000 | 2.005 |
|                                                                      | 1.87 mW cm <sup>-2</sup>             | 1.983 | 1.993 | 2.003 |
|                                                                      | 3.74 mW cm <sup>-2</sup>             | 1.998 | 2.001 | 1.996 |
|                                                                      | 5.0 mW cm <sup>-2</sup>              | 2.001 | 1.987 | 2.005 |
|                                                                      | 6.25 mW cm <sup>-2</sup>             | 1.991 | 1.988 | 1.990 |

## References

1. A. Kudo, Y. Miseki, Heterogeneous photocatalyst materials for water splitting. *Chem. Soc. Rev.* **38**, 253–278 (2009). [Medline doi:10.1039/b800489g](#)
2. A. Iwase, Y. H. Ng, Y. Ishiguro, A. Kudo, R. Amal, Reduced graphene oxide as a solid-state electron mediator in Z-scheme photocatalytic water splitting under visible light. *J. Am. Chem. Soc.* **133**, 11054–11057 (2011). [Medline doi:10.1021/ja203296z](#)
3. B. A. Pinaud, J. D. Benck, L. C. Seitz, A. J. Forman, Z. Chen, T. G. Deutsch, B. D. James, K. N. Baum, G. N. Baum, S. Ardo, H. Wang, E. Miller, T. F. Jaramillo, Technical and economic feasibility of centralized facilities for solar hydrogen production via photocatalysis and photoelectrochemistry. *Energy Environ. Sci.* **6**, 1983–2002 (2013). [doi:10.1039/c3ee40831k](#)
4. T. J. Jacobsson, V. Fjällström, M. Sahlberg, M. Edoff, T. Edvinsson, A monolithic device for solar water splitting based on series interconnected thin film absorbers reaching over 10% solar-to-hydrogen efficiency. *Energy Environ. Sci.* **6**, 3676–3683 (2013). [doi:10.1039/c3ee42519c](#)
5. J. Luo, J. H. Im, M. T. Mayer, M. Schreier, M. K. Nazeeruddin, N. G. Park, S. D. Tilley, H. J. Fan, M. Grätzel, Water photolysis at 12.3% efficiency via perovskite photovoltaics and Earth-abundant catalysts. *Science* **345**, 1593–1596 (2014). [Medline doi:10.1126/science.1258307](#)
6. S. Licht, Multiple band gap semiconductor/electrolyte solar energy conversion. *J. Phys. Chem. B* **105**, 6281–6294 (2001). [doi:10.1021/jp010552j](#)
7. M. S. Prévot, K. Sivula, Photoelectrochemical tandem cells for solar water splitting. *J. Phys. Chem. C* **117**, 17879–17893 (2013). [doi:10.1021/jp405291g](#)
8. K. Domen, S. Naito, M. Soma, T. Onishi, K. J. Tamaru, Photocatalytic decomposition of water vapor on an NiO-SrTiO<sub>3</sub> catalyst. *J. Chem. Soc. Chem. Commun.* **12**, 543–544 (1980). [doi:10.1039/c39800000543](#)
9. A. Kudo, Y. Miseki, Heterogeneous photocatalyst materials for water splitting. *Chem. Soc. Rev.* **38**, 253–278 (2009). [Medline doi:10.1039/b800489g](#)

10. W. J. Youngblood, S. H. A. Lee, K. Maeda, T. E. Mallouk, Visible light water splitting using dye-sensitized oxide semiconductors. *Acc. Chem. Res.* **42**, 1966–1973 (2009). [Medline doi:10.1021/ar9002398](#)
11. K. Maeda, K. Domen, Photocatalytic water splitting: Recent progress and future challenges. *J. Phys. Chem. Lett.* **1**, 2655–2661 (2010). [doi:10.1021/jz1007966](#)
12. X. Chen, S. Shen, L. Guo, S. S. Mao, Semiconductor-based photocatalytic hydrogen generation. *Chem. Rev.* **110**, 6503–6570 (2010). [Medline doi:10.1021/cr1001645](#)
13. K. Maeda, Z-scheme water splitting using two different semiconductor photocatalysts. *ACS Catal.* **3**, 1486–1503 (2013). [doi:10.1021/cs4002089](#)
14. K. Maeda, K. Teramura, D. Lu, T. Takata, N. Saito, Y. Inoue, K. Domen, Characterization of Rh-Cr mixed-oxide nanoparticles dispersed on  $(\text{Ga}_{1-x}\text{Zn}_x)(\text{N}_{1-x}\text{O}_x)$  as a cocatalyst for visible-light-driven overall water splitting. *J. Phys. Chem. B* **110**, 13753–13758 (2006). [Medline doi:10.1021/jp061829o](#)
15. K. Maeda, K. Teramura, D. Lu, T. Takata, N. Saito, Y. Inoue, K. Domen, Photocatalyst releasing hydrogen from water. *Nature* **440**, 295 (2006). [Medline doi:10.1038/440295a](#)
16. V. Balzani, A. Credi, M. Venturi, Photochemical conversion of solar energy. *ChemSusChem* **1**, 26–58 (2008). [Medline doi:10.1002/cssc.200700087](#)
17. H. B. Gray, Powering the planet with solar fuel. *Nat. Chem.* **1**, 7 (2009). [Medline doi:10.1038/nchem.141](#)
18. I. Tsuji, H. Kato, A. Kudo, Visible-light-induced  $\text{H}_2$  evolution from an aqueous solution containing sulfide and sulfite over a  $\text{ZnS-CuInS}_2\text{-AgInS}_2$  solid-solution photocatalyst. *Angew. Chem. Int. Ed.* **44**, 3565–3568 (2005). [Medline doi:10.1002/anie.200500314](#)
19. J. Liu, Y. Zhang, L. Lu, G. Wu, W. Chen, Self-regenerated solar-driven photocatalytic water-splitting by urea derived graphitic carbon nitride with platinum nanoparticles. *Chem. Commun.* **48**, 8826–8828 (2012). [Medline doi:10.1039/c2cc33644h](#)
20. F. Rappaport, M. Guergova-Kuras, P. J. Nixon, B. A. Diner, J. Lavergne, Kinetics and pathways of charge recombination in photosystem II. *Biochemistry* **41**, 8518–8527 (2002). [Medline doi:10.1021/bi025725p](#)

21. G. Ananyev, G. C. Dismukes, How fast can photosystem II split water? Kinetic performance at high and low frequencies. *Photosynth. Res.* **84**, 355–365 (2005). [Medline](#) [doi:10.1007/s11120-004-7081-1](https://doi.org/10.1007/s11120-004-7081-1)
22. D. J. Martin, P. J. Reardon, S. J. Moniz, J. Tang, Visible light-driven pure water splitting by a nature-inspired organic semiconductor-based system. *J. Am. Chem. Soc.* **136**, 12568–12571 (2014). [Medline](#) [doi:10.1021/ja506386e](https://doi.org/10.1021/ja506386e)
23. F. He, G. Chen, Y. Yu, S. Hao, Y. Zhou, Y. Zheng, Facile approach to synthesize g-PAN/g-C<sub>3</sub>N<sub>4</sub> composites with enhanced photocatalytic H<sub>2</sub> evolution activity. *ACS Appl. Mater. Interfaces* **6**, 7171–7179 (2014). [Medline](#) [doi:10.1021/am500198y](https://doi.org/10.1021/am500198y)
24. X. Zhang, L. Yu, C. Zhuang, T. Peng, R. Li, X. Li, Highly Asymmetric phthalocyanine as a sensitizer of graphitic carbon nitride for extremely efficient photocatalytic H<sub>2</sub> production under near-infrared light. *ACS Catal.* **4**, 162–170 (2014). [doi:10.1021/cs400863c](https://doi.org/10.1021/cs400863c)
25. K. Schwinghammer, M. B. Mesch, V. Duppel, C. Ziegler, J. Senker, B. V. Lotsch, Crystalline carbon nitride nanosheets for improved visible-light hydrogen evolution. *J. Am. Chem. Soc.* **136**, 1730–1733 (2014). [Medline](#) [doi:10.1021/ja411321s](https://doi.org/10.1021/ja411321s)
26. R. Marschall, Semiconductor composites: Strategies for enhancing charge carrier separation to improve photocatalytic activity. *Adv. Mater.* **24**, 2421–2440 (2014).
27. J. Liebig, About some nitrogen compounds. *Ann. Pharm.* **10**, 10 (1834).
28. S. Yin, Y. Aita, M. Komatsu, T. Sato, Visible-light-induced photocatalytic activity of TiO<sub>2-x</sub>N<sub>y</sub> prepared by solvothermal process in urea-alcohol system. *J. Eur. Ceram. Soc.* **26**, 2735–2742 (2006). [doi:10.1016/j.jeurceramsoc.2005.05.012](https://doi.org/10.1016/j.jeurceramsoc.2005.05.012)
29. X. Wang, K. Maeda, A. Thomas, K. Takanabe, G. Xin, J. M. Carlsson, K. Domen, M. Antonietti, A metal-free polymeric photocatalyst for hydrogen production from water under visible light. *Nat. Mater.* **8**, 76–80 (2009). [Medline](#) [doi:10.1038/nmat2317](https://doi.org/10.1038/nmat2317)
30. B. Chai, T. Peng, J. Mao, K. Li, L. Zan, Graphitic carbon nitride (g-C<sub>3</sub>N<sub>4</sub>)-Pt-TiO<sub>2</sub> nanocomposite as an efficient photocatalyst for hydrogen production under visible light irradiation. *Phys. Chem. Chem. Phys.* **14**, 16745–16752 (2012). [Medline](#) [doi:10.1039/c2cp42484c](https://doi.org/10.1039/c2cp42484c)

31. J. Hong, Y. Wang, Y. Wang, W. Zhang, R. Xu, Noble-metal-free NiS/C<sub>3</sub>N<sub>4</sub> for efficient photocatalytic hydrogen evolution from water. *ChemSusChem* **6**, 2263–2268 (2013). [Medline doi:10.1002/cssc.201300647](#)
32. Q. J. Xiang, J. G. Yu, M. Jaroniec, Preparation and enhanced visible-light photocatalytic H<sub>2</sub>-production activity of graphene/C<sub>3</sub>N<sub>4</sub> composites. *J. Phys. Chem. C* **115**, 7355–7363 (2011). [doi:10.1021/jp200953k](#)
33. A. Suryawanshi, P. Dhanasekaran, D. Mhamane, S. Kelkar, S. Patil, N. Gupta, S. Ogale, Doubling of photocatalytic H<sub>2</sub> evolution from g-C<sub>3</sub>N<sub>4</sub> via its nanocomposites formation with multiwall carbon nanotubes: Electronic and morphological effects. *Int. J. Hydrogen Energy* **37**, 9584–9589 (2012). [doi:10.1016/j.ijhydene.2012.03.123](#)
34. S. Yang, Y. Gong, J. Zhang, L. Zhan, L. Ma, Z. Fang, R. Vajtai, X. Wang, P. M. Ajayan, Exfoliated graphitic carbon nitride nanosheets as efficient catalysts for hydrogen evolution under visible light. *Adv. Mater.* **25**, 2452–2456 (2013). [Medline doi:10.1002/adma.201204453](#)
35. L. Yu, X. Zhang, C. Zhuang, L. Lin, R. Li, T. Peng, Syntheses of asymmetric zinc phthalocyanines as sensitizer of Pt-loaded graphitic carbon nitride for efficient visible/near-IR-light-driven H<sub>2</sub> production. *Phys. Chem. Chem. Phys.* **16**, 4106–4114 (2014). [Medline doi:10.1039/c3cp54316a](#)
36. H. T. Li, Z. H. Kang, Y. Liu, S. T. Lee, Carbon nanodots: Synthesis, properties and applications. *J. Mater. Chem.* **22**, 24230–24253 (2012). [doi:10.1039/c2jm34690g](#)
37. H. Ming, Z. Ma, Y. Liu, K. Pan, H. Yu, F. Wang, Z. Kang, Large scale electrochemical synthesis of high quality carbon nanodots and their photocatalytic property. *Dalton Trans.* **41**, 9526–9531 (2012). [Medline doi:10.1039/c2dt30985h](#)
38. X. Wang, K. Maeda, X. Chen, K. Takanabe, K. Domen, Y. Hou, X. Fu, M. Antonietti, Polymer semiconductors for artificial photosynthesis: Hydrogen evolution by mesoporous graphitic carbon nitride with visible light. *J. Am. Chem. Soc.* **131**, 1680–1681 (2009). [Medline doi:10.1021/ja809307s](#)

39. M. Yang, Q. Huang, X. Q. Jin, ZnGaNO solid solution-C<sub>3</sub>N<sub>4</sub> composite for improved visible light photocatalytic performance. *Mater. Sci. Eng. B* **177**, 600–605 (2012).  
[doi:10.1016/j.mseb.2012.03.004](https://doi.org/10.1016/j.mseb.2012.03.004)
40. L. Liao, Q. Zhang, Z. Su, Z. Zhao, Y. Wang, Y. Li, X. Lu, D. Wei, G. Feng, Q. Yu, X. Cai, J. Zhao, Z. Ren, H. Fang, F. Robles-Hernandez, S. Baldelli, J. Bao, Efficient solar water-splitting using a nanocrystalline CoO photocatalyst. *Nat. Nanotechnol.* **9**, 69–73 (2014). [Medline](https://pubmed.ncbi.nlm.nih.gov/25411111/) [doi:10.1038/nnano.2013.272](https://doi.org/10.1038/nnano.2013.272)
41. S. Mubeen, J. Lee, N. Singh, S. Krämer, G. D. Stucky, M. Moskovits, An autonomous photosynthetic device in which all charge carriers derive from surface plasmons. *Nat. Nanotechnol.* **8**, 247–251 (2013). [Medline](https://pubmed.ncbi.nlm.nih.gov/24011111/) [doi:10.1038/nnano.2013.18](https://doi.org/10.1038/nnano.2013.18)
42. R. W. Baker, Future directions of membrane gas separation technology. *Ind. Eng. Chem. Res.* **41**, 1393–1411 (2002). [doi:10.1021/ie0108088](https://doi.org/10.1021/ie0108088)
43. Y. J. Ando, T. Tanaka, Proposal for a new system for simultaneous production of hydrogen and hydrogen peroxide by water electrolysis. *Int. J. Hydrogen Energy* **29**, 1349–1354 (2004).  
[doi:10.1016/j.ijhydene.2004.02.001](https://doi.org/10.1016/j.ijhydene.2004.02.001)
44. X. Chen, Y.-S. Jun, K. Takanabe, K. Maeda, K. Domen, X. Fu, M. Antonietti, X. Wang, Ordered mesoporous SBA-15 type graphitic carbon nitride: A semiconductor host structure for photocatalytic hydrogen evolution with visible light. *Chem. Mater.* **21**, 4093–4095 (2009). [doi:10.1021/cm902130z](https://doi.org/10.1021/cm902130z)
45. J. H. Liu, T. K. Zhang, Z. C. Wang, G. Dawson, W. Chen, Simple pyrolysis of urea into graphitic carbon nitride with recyclable adsorption and photocatalytic activity. *J. Mater. Chem.* **21**, 14398–14401 (2011). [doi:10.1039/c1jm12620b](https://doi.org/10.1039/c1jm12620b)
46. E. Quirico, G. Montagnac, V. Lees, P. F. McMillan, C. Szopa, G. Cernogora, J.-N. Rouzaud, P. Simon, J.-M. Bernard, P. Coll, N. Fray, R. D. Minard, F. Raulin, B. Reynard, B. Schmitt, New experimental constraints on the composition and structure of tholins. *Icarus* **198**, 218–231 (2008). [doi:10.1016/j.icarus.2008.07.012](https://doi.org/10.1016/j.icarus.2008.07.012)
47. M. J. Bojdys, J. O. Müller, M. Antonietti, A. Thomas, Ionothermal synthesis of crystalline, condensed, graphitic carbon nitride. *Chem. Eur. J.* **14**, 8177–8182 (2008). [Medline](https://pubmed.ncbi.nlm.nih.gov/18411111/) [doi:10.1002/chem.200800190](https://doi.org/10.1002/chem.200800190)

48. A. Y. Liu, M. L. Cohen, Prediction of new low compressibility solids. *Science* **245**, 841–842 (1989). [Medline doi:10.1126/science.245.4920.841](#)
49. V. N. Khabashesku, J. L. Zimmerman, J. L. Margrave, Powder synthesis and characterization of amorphous carbon nitride. *Chem. Mater.* **12**, 3264–3270 (2000). [doi:10.1021/cm000328r](#)
50. T. Komatsu, T. Nakamura, Polycondensation/pyrolysis of tris-s-triazine derivatives leading to graphite-like carbon nitrides. *J. Mater. Chem.* **11**, 474–478 (2001). [doi:10.1039/b005982j](#)
51. Q. Guo, Y. Xie, X. Wang, S. Zhang, T. Hou, S. Lv, Synthesis of carbon nitride nanotubes with the C<sub>3</sub>N<sub>4</sub> stoichiometry via a benzene-thermal process at low temperatures. *Chem. Commun.* **2004**, 26–27 (2004). [Medline doi:10.1039/b311390f](#)
52. A. Thomas, A. Fischer, F. Goettmann, M. Antonietti, J.-O. Müller, R. Schlögl, J. M. Carlsson, Graphitic carbon nitride materials: Variation of structure and morphology and their use as metal-free catalysts. *J. Mater. Chem.* **18**, 4893–4908 (2008). [doi:10.1039/b800274f](#)
53. C. Liu, P. Zhang, X. Zhai, F. Tian, W. Li, J. Yang, Y. Liu, H. Wang, W. Wang, W. Liu, Nano-carrier for gene delivery and bioimaging based on carbon dots with PEI-passivation enhanced fluorescence. *Biomaterials* **33**, 3604–3613 (2012). [Medline doi:10.1016/j.biomaterials.2012.01.052](#)
54. R. H. Liu, H. Huang, H. Li, Y. Liu, J. Zhong, Y. Li, S. Zhang, Z. Kang, Metal nanoparticle/carbon quantum dot composite as photocatalysts for high efficiency cyclohexane oxidation. *ACS Catal.* **4**, 328–336 (2014). [doi:10.1021/cs400913h](#)